Supporting Information for

Short communication

Development of an active-site titrant for SARS-CoV-2 main protease as an indispensable tool for evaluating enzyme kinetics

Rabea Voget[a], Julian Breidenbach[a,†], Tobias Claff[a,†], Alexandra Hingst[a,†], Katharina Sylvester[a], Christian Steinebach[a], Lan Phuong Vu[a], Renato H. Weiße[b], Ulrike Bartz[c], Norbert Sträter[b], Christa E. Müller[a], Michael Gütschow[a,*]

[a]*Pharmaceutical Institute, Pharmaceutical & Medicinal Chemistry, University of Bonn, Bonn 53121, Germany*

[b]*Institute of Bioanalytical Chemistry, Center for Biotechnology and Biomedicine, Leipzig University, Leipzig 04103, Germany*

[c]*Department of Natural Sciences, University of Applied Sciences Bonn-Rhein-Sieg, Rheinbach 53359, Germany*

[†]These authors made equal contributions to this work.

*Corresponding author.

E-mail address: guetschow@uni-bonn.de (Michael Gütschow).

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1. Biochemistry

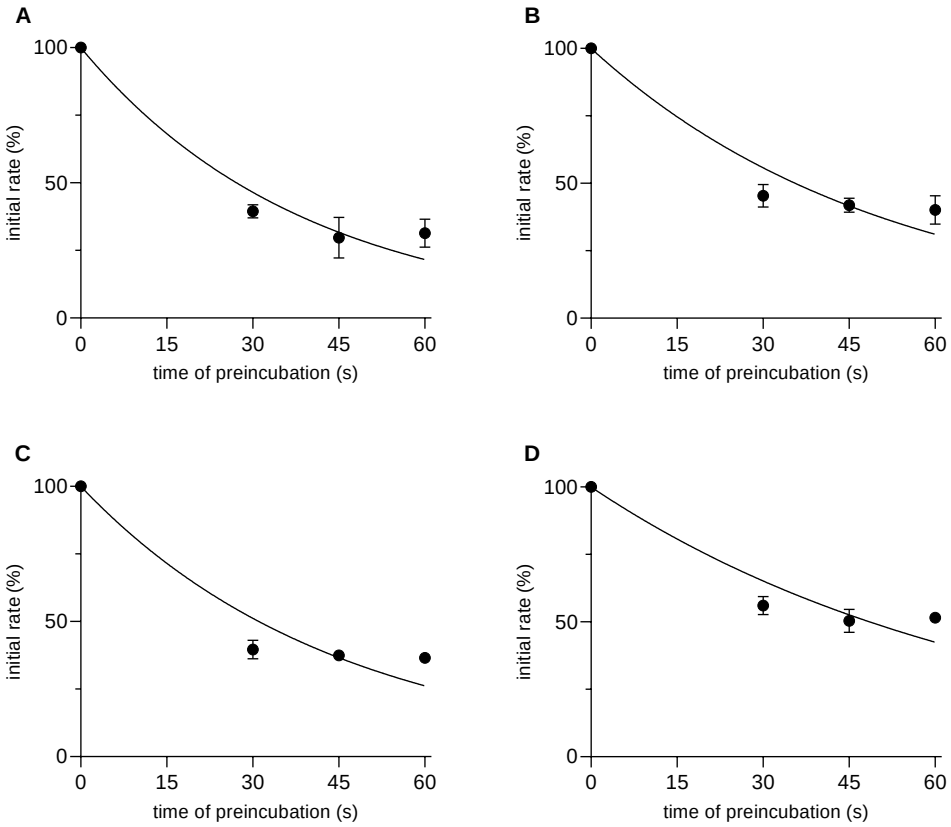


Figure S1. Plot of the residual initial rates *versus* pre-incubation time to determine the second-order rate constant $k_{obs}/[I]$ for M^{pro} inactivation by **1**. Four different substrates were employed, i.e., 50 µmol/L of **2** (**A**), 200 µmol/L of **3** (**B**) and **4** (**C**), and 50 µmol/L of **5** (**D**). $k_{obs}/[I]$ values for **1** under use of **2**, **3**, **4** and **5** were determined as 128000 L/mol·s, 97400 L/mol·s, 112000 L/mol·s and 71400 L/mol·s, respectively. Data are means from triplicate measurements.

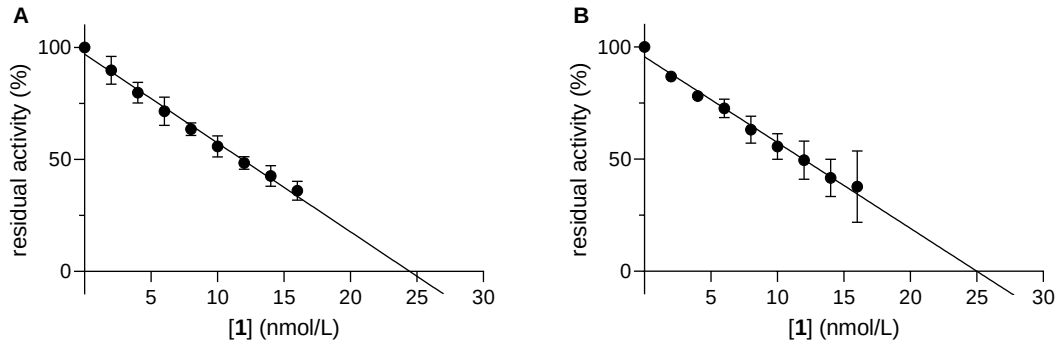


Figure S2. Active-site titration of His-tagged SARS-CoV-2 M^{pro} (1.64 ng/µL) with **1** under use of substrate **2**. After a pre-incubation time of 5 min (**A**) or 20 min (**B**), respectively, **2** was added to a final concentration of 50 µmol/L and the residual enzyme activity was determined over 10 min. For further experimental details, see Materials and methods in the main body of the manuscript. The obtained active-site concentrations were determined as 24.5 ± 0.7 nmol/L (**A**) or 25.0 ± 1.1 nmol/L (**B**), respectively. The data are means from three independent experiments performed with aliquots from the same enzyme preparation. The standard deviations refer to the linear regression.

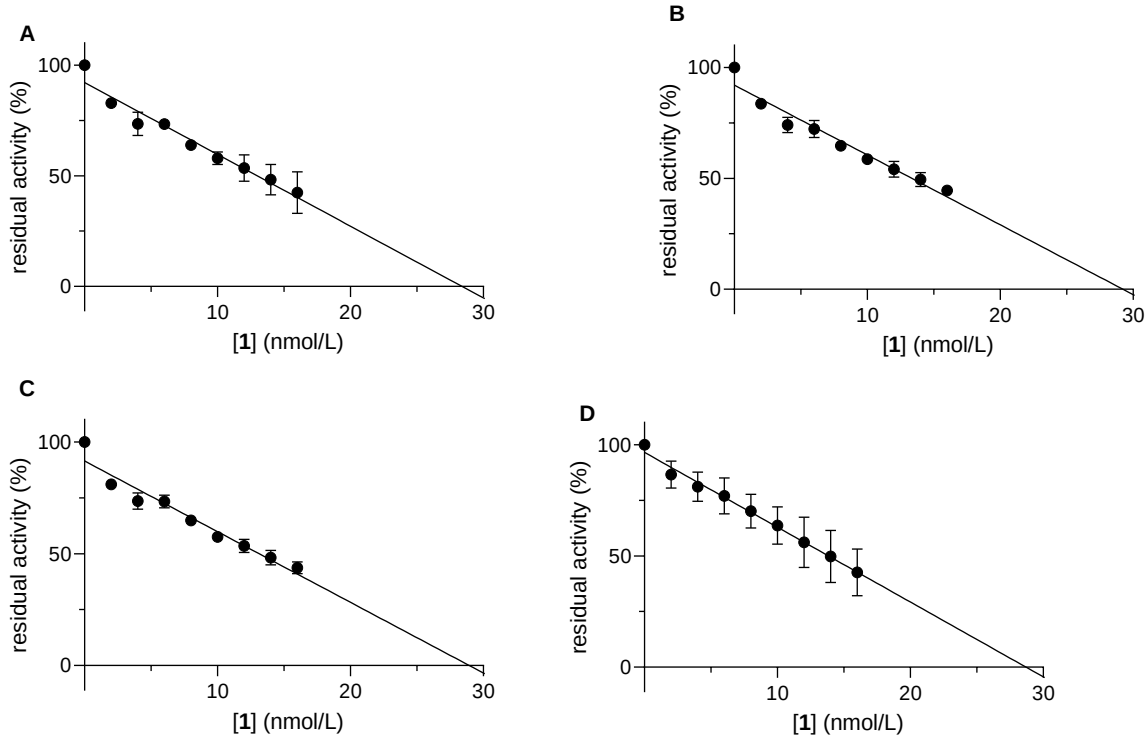


Figure S3. Active-site titration of His-tagged SARS-CoV-2 M^{pro} (1.64 ng/µL) with **1** under use of substrate **2** (**A**), **3** (**B**), **4** (**C**), and **5** (**D**), for structures see Fig. S8. After a pre-incubation time of 20 min, substrate **2** (50 µmol/L) or **3** (200 µmol/L) or **4** (200 µmol/L) or **5** (25 µmol/L) was added and the residual enzyme activity was determined over 10 min. For further experimental details, see Materials and methods in the main body of the manuscript. The obtained active-site concentrations were determined as 28.4 ± 1.4 nmol/L (**A**), 29.2 ± 1.1 nmol/L (**B**), 28.9 ± 1.2 nmol/L (**C**), and 28.7 ± 1.6 nmol/L (**D**). The data are means from three independent experiments performed with aliquots from the same enzyme preparation. The standard deviations refer to the linear regression.

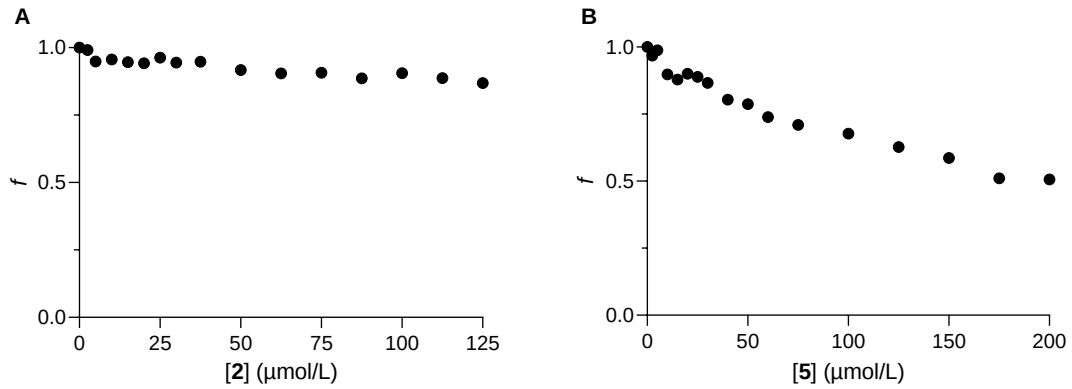


Figure S4. Dependency of the correction factor *f* on the concentration of substrate **2** (**A**) and substrate **5** (**B**).

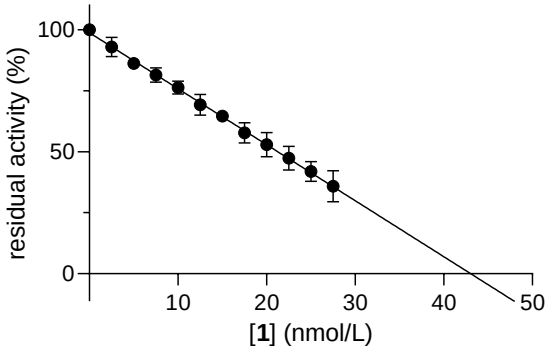


Figure S5. Active-site titration of purified native SARS-CoV-2 M^{pro} (7.5 ng/µL) with **1** under use of substrate **2** (50 µmol/L). The data are means from three independent experiments performed with aliquots from the same enzyme preparation. The equivalent point, obtained by linear regression as the x-axis intercept, corresponded to an active enzyme concentration of 43.0 ± 0.9 nmol/L and a proportion of 20% active enzyme. The standard deviations refer to the linear regression.

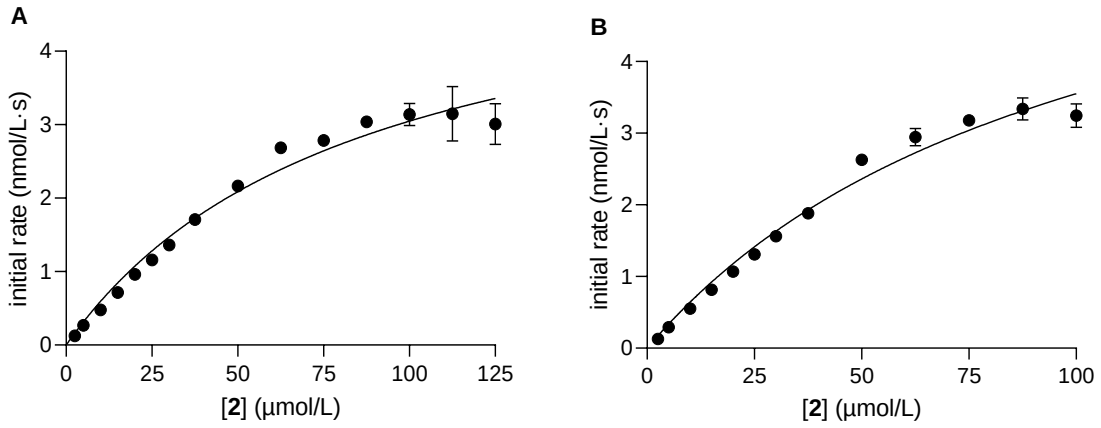


Figure S6. Determination of k_{cat}, k_{cat}/K_m, and K_m for the conversion of substrate **2** by His-tagged and native M^{pro}. Data are means from triplicate measurements. Fluorescence values at each concentration were corrected by the IFE by means of experimentally determined correction factors. (A) The C-terminally His-tagged protease was used (2 ng/µL with a fraction of 56% of active enzyme). Data are also shown in the main body of the manuscript. The following kinetic parameters were obtained k_{cat} = 0.173 s^{-1}, k_{cat}/K_m = 2030 L/mol·s, K_m = 85.2 ± 5.6 µmol/L. (B) The native M^{pro} was used (5.65 ng/µL with a fraction of 20% of active enzyme). The following kinetic parameters were obtained k_{cat} = 0.221 s^{-1}, k_{cat}/K_m = 2170 L/mol·s, K_m = 102 ± 13 µmol/L.

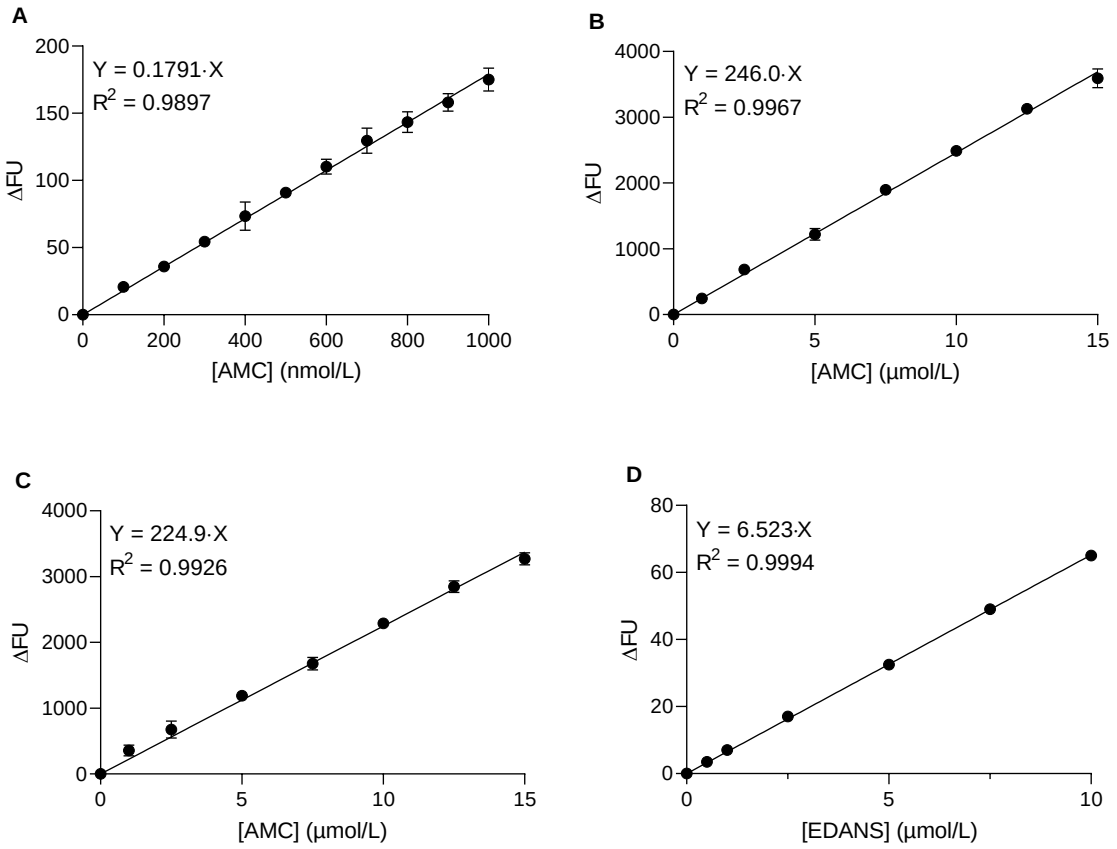


Figure S7. Fluorescence *versus* product concentration calibration lines. **A**. Calibration for **2** on the basis of free AMC; **B**-**C**. Lines quantifying the change in fluorescence readout resulting from different concentrations of AMC in the presence of the respective substrate; **B.** Calibration at [**3**] = 3.48 mmol/L; **C.** Calibration at [**4**] = 3.30 mmol/L; **D.** Calibration for **5** on the basis of free EDANS. All data are means from duplicate measurements.

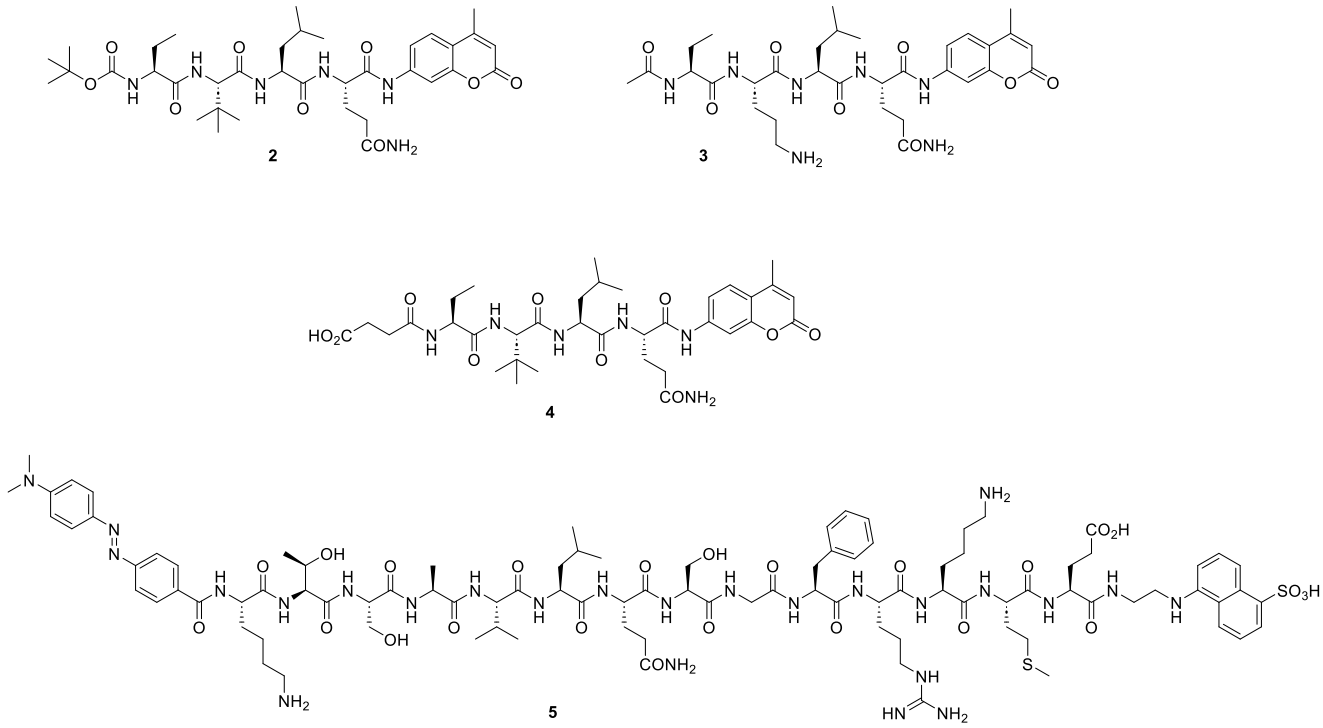


Figure S8. Chemical structures of substrates **2**-**5**.

2. Determination of Substrate Solubility

Substrate solubility was determined by the shake flask method using HPLC analysis. About 1 mg (substrates **2, 5**) or 1.25 mg (substrates **3**, **4**) was weighed into a reaction tube. A volume of 1 mL (**2**, **5**) or 250 µL (**3**, **4**) of assay buffer (50 mmol/L MOPS, 10 mmol/L NaCl, 1 mmol/L EDTA and 0.01% Triton X-100, pH 7.2) containing 5% (*v*/*v*) DMSO was added. Subsequently, the suspensions were shaken for 1 h with 450 rpm and 37 °C and then centrifuged at 15,000 rpm for 10 min. Aliquots of 450 µL (**2**) or 50 µL (**3**, **4**, **5**) of the supernatants were removed and diluted with 50 µL (**2**) or 950 µL (**3**, **4**, **5**) of acetonitrile to prevent precipitation of the dissolved compounds and to obtain sufficient volume for the measurement. Substrate **2** was analyzed on a Poroshell 120 EC-C18 3 × 50 mm, 2.7 µm column with an injection volume of 20 µL and a flow rate of 0.7 mL/min. The substrates **3**, **4** and **5** were analyzed on a Nucleosil EC 4 × 250 mm, 5 µm column with an injection volume of 10 µL (**3**, **4**) or 20 µL (**5**) and a flow rate of 1.0 mL/min. Elution occurred within 12 min (**2**) or 15 min (**3**, **4**, **5**) in a linear gradient between eluent A consisting of 0.1% TFA in water, and eluent B consisting of 0.1% TFA in acetonitrile. The starting concentration of eluent B was 20% (**2**) or 30% (**3**, **4**, **5**) and the final concentration was 80% (**2**) or 95% (**3**, **4**, **5**). UV detection was carried out at 270 nm (**2**, **3**, **4**) or 490 nm (**5**). For the determination of the sample concentrations, calibration lines for each substrate were obtained. The respective compound was accurately weighed and a DMSO stock solution of 1 mg/mL was prepared. The solution was diluted with acetonitrile to yield six different concentrations between 1 and 300 µg/mL, which were analyzed by HPLC. The peak area was plotted against the calibration solution concentration and thus sample concentrations could be calculated under consideration of the dilution of the supernatants with acetonitrile.

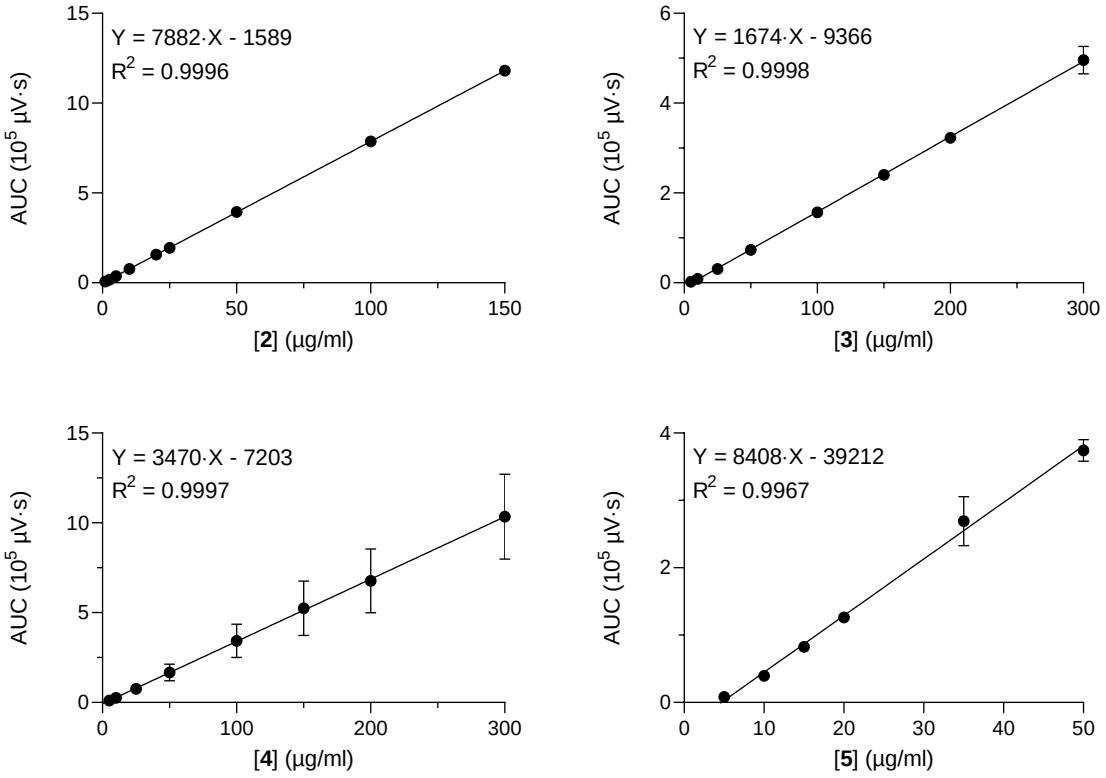


Figure S9. Calibration lines for HPLC-based determination of substrate solubility. All data are means from duplicate measurements.

3. X-ray Crystallography

E. coli strain BL21(DE3) was transformed with the sequence-verified SARS-CoV-2 M^{pro} construct as previously described[1]. Transformed clones were pre-cultured at 37 °C in 100 mL LB-medium with ampicillin (100 µg/mL) overnight. The next morning, the incubated culture was inoculated into 2 L of LB-medium supplied with 100 µg/mL ampicillin. 1 mmol/L isopropyl-D-thiogalactoside (IPTG) was added for induction of the M^{pro} gene at 37 °C when the OD_{600} reached 0.5. After 4 h, cells were harvested by centrifugation at 10,000 × *g* and 4 °C for 10 min. The pellet was either frozen at -80 °C or directly processed further. The pellet was resuspended in 20 mL of buffer A (20 mmol/L Tris, 150 mmol/L NaCl, pH 7.8; pH values of all buffers were adjusted at room temperature) and then lyzed by repeated sonication. The lysate was centrifuged at 48,000 × *g* and 4 °C for 30 min. The supernatant was loaded onto a HiTrap Talon crude 5 mL column (Cytiva) equilibrated with buffer A. The column was washed with 10 column volumes of buffer A to remove nonspecifically binding proteins, followed by elution using buffer B (20 mmol/L Tris, 150 mmol/L NaCl, 500 mmol/L imidazole, pH 7.8) applying a linear gradient of imidazole ranging from 0 to 250 mmol/L within 30 column volumes. The fractions containing the target protein were pooled and mixed with human rhinovirus (HRV) 3C protease (ThermoFisher) at a concentration of 5 U/mg protein and dialyzed into buffer C (20 mmol/L Tris, 150 mmol/L NaCl, 1 mmol/L tris(2-carboxyethyl)phosphine (TCEP), pH 7.8) at 4 °C overnight. The HRV 3C protease-treated M^{pro} was again applied to a connected HiTrap Talon crude 5 mL column (Cytiva) to remove the His-tagged HRV 3C protease, the His-tag, and protein with uncleaved His-tag. The His-tag-free M^{pro} in the flow-through was subjected to buffer exchange with buffer D (20 mmol/L 2-[4-(2-hydroxyethyl)piperazin-1-yl]ethanesulfonic acid (HEPES), 1 mmol/L TCEP, pH 7.4) using Amicon Ultra 15 centrifugal filters (10 kDa molecular weight cut-off, Merck Millipore) at 5000 × *g*, and 4 °C.

Freshly prepared protein (as described above) at a concentration of 3 mg/mL (around 100 µmol/L) was mixed with compound **1** (dissolved in 100% DMSO) at a molar ratio of 1:2. The mixture was incubated at room temperature for 1 h and then centrifuged at 12,000 × *g*. Subsequently, a basic screen with the commercially available Morpheus HT (Molecular Dimensions) was performed by using a Formulatrix NT8-Drop Setter employing the sitting-drop vapor-diffusion method at 18 °C. Protein solution (0.15 µL) and 0.15 µL of reservoir were mixed to equilibrate against 40 µL of reservoir solution containing 200 µmol/L of titrant **1**. Crystals started appearing after 48 h applying the following conditions: 0.1 mol/L HEPES/3-(*N*-morpholino)propanesulfonic acid (MOPS) (pH 7.5, 20 °C), 30 mmol/L sodium fluoride, 30 mmol/L sodium bromide, 30 mmol/L sodium iodide, 12% (*v*/*v*) polyethylene glycol (PEG) methyl ether 500, 6% (*w*/*v*) PEG20000. Subsequently, crystals were harvested and flash-frozen in liquid nitrogen without additional cryo-protection.

X-ray diffraction data were collected at 100 K at beamline P14 at the DESY synchrotron in Hamburg, Germany (Table S1). The diffraction data were indexed, integrated and scaled with XDS[2] and STARANISO[3] as implemented in ISPyB[4] at DESY. The obtained crystal form has so far only been observed for a complex structure of SARS-CoV-1 main protease (PDB ID 1UJ1)[5], which was used as a starting model for refinement in Phenix[6]. Coot[7] was used for model building. The structure contains two copies in the asymmetric unit, which were refined with non-crystallographic symmetry restraints on the torsion angles. The difference density indicated binding of the ligand in both proteins of the asymmetric unit. Stereochemical restraints for ligand refinement were generated

using grade (https://grade.globalphasing.org). In chain A of the asymmetric unit, the average B-factor of the ligand (54.0 Å^2) was significantly higher compared to chain B (23.6 Å^2). A similar difference was observed for the average B-factor of the surrounding amino acids for chain A (42.9 Å^2) and chain B (26.0 Å^2). The average B-factors of the complete proteins were similar with B = 32.4 Å^2 for chain A and 31.9 Å^2 for chain B. The lower B-factors of the ligand and its environment in chain B is likely a result of crystal packing interactions of the Cbz capping group of the ligand and residues of the neighboring loops. This is in contrast to chain A, where the protease active site and the ligand are situated at more than 8 Å distance to symmetry mates in the crystal structure. Nevertheless, a superposition of both chains shows very similar binding modes of the ligands in the two chains (Fig. S10B). The terminal benzyl moiety of **1**, part of the Cbz group, lacks sufficient electron density and is presumed to be flexible. Overall, the B-factors and the electron density of **1** suggest full occupancy of the ligand.

Table S1. Diffraction data and refinement statistics

Compound	**1**
PDB entry ID	8QDC
source	DESY PETRA III EMBL beamline P14
wavelength (Å)	0.97625
Resolution (Å)	54.51-1.77 (2.00-1.77)
Resolution aniso (Å)	2.095, 2.382 (b*), 1.755
Space group	$P2_1$
Unit cell dimensions (Å,°)	54.01, 98.65, 67.53, β = 104.4
Unique reflections	39350 (1968)
Multiplicity	7.0 (7.2)
Completeness (%) spherical/ellipsoidal	59.1 (9.6) / 92.2 (64.8)
Mean I/σ (I)	12.4 (1.8)
R_{merge}	0.082 (1.013)
R_{pim}	0.034 (0.404)
$CC_{1/2}$	0.998 (0.672)
Wilson B-factor ($Å^2$)	26.8
Resolution refinement (Å)	54.51-1.77 (1.81-1.77)
R-work	0.1940 (0.3327)
R-free	0.2391 (0.1606)[1]
Number of non-hydrogen atoms, B-factor ($Å^2$)	
protein	4791 (35.3)
ligand	94 (42.8)
solvent	295 (34.8)
Rmsd bonds (Å)	0.006
Rmsd angles (°)	0.84
Ramachandran favored (%)	98.7
Ramachandran allowed (%)	1.34
Ramachandran outliers (%)	0.0
Rotamer outliers (%)	2.62
MolProbity clash score	2.28

[1]Due to anisotropic truncation, the test set of the highest resolution shell contains only one reflection. The R-free value is 0.3829 in the shell 1.86-1.82 Å.

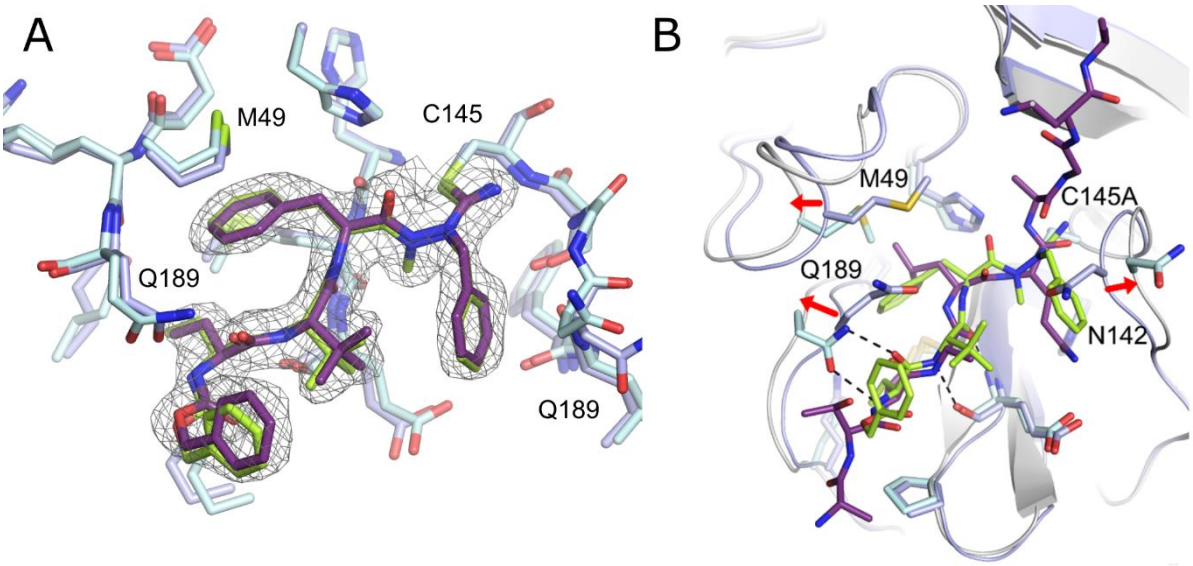


Figure S10. Comparison of ligand binding modes. (A) Comparison of the binding mode of **1** to chain A (ligand green, protein cyan) and chain B (ligand purple, protein blue). The ($2F_o$-F_c)-type electron density of the ligand in chain B is shown at a contour level of 1.0 σ_{rms}. (B) Superposition of cocrystal structures of M^{pro} (cyan) with **1** (green, PDB ID 8QDC) and a decapeptide substrate (ATVRLQAGNA with unresolved arginine side chain; purple) bound to a C145A variant (blue, PDB ID 7TA4)[8]. The surface loops comprising M49, N142, and Q189 show significant rearrangements (illustrated by red arrows), which enable the interaction of **1** with M^{pro}. M49 and Q189 need to reposition for the hydrogen bonding interaction of the side chain amide of Q189 with the P4 amino acid of the inhibitor. N142 changes conformation as the phenyl ring of the P1 building block does not bind into the deep S1 pocket but is located above this pocket.

4. Synthetic Chemistry

4.1. General Procedures and Instruments

Chemicals, reagents and solvents were purchased from abcr GmbH (Karlsruhe, Germany), Sigma-Aldrich Chemie GmbH (Taufkirchen, Germany), TCI Deutschland GmbH (Eschborn, Germany), BLD Pharmatech GmbH (Kaiserslautern, Germany), Fisher Scientific GmbH (Schwerte) and JULI.O GmbH (Jülich). Thin-layer chromatography was conducted on silica gel 60 F_{254} aluminum sheets (Merck KGaA, Darmstadt) and detected at 254 nm. Retardation factors (R_f) are quoted. Melting points are determined on a Büchi (Essen, Germany) M-560 apparatus. ^{1}H NMR (500 MHz) and ^{13}C NMR (126 MHz) spectra were recorded on a Bruker (Rheinstetten, Germany) DRX 500 MHz Avance and ^{1}H NMR (600 MHz) and ^{13}C NMR (151 MHz) spectra on a Bruker Ascend 600 at 300 – 303 K. Chemical shift values δ are given in ppm and are referenced to solvent signals, DMSO-d_6 2.50/39.70 ppm. Flash chromatography was carried out at Interchim SAS (Montluçon, France) puriFlash XS520Plus. LC-MS analyses (method A) were performed on an Agilent (Santa Clara, CA, United States) HPLC 1260 Infinity II using an EC50/2 Nucleodur C18 Gravity 3 µm column (Macherey-Nagel, Düren, Germnany) coupled to an Agilent (Santa Clara, CA, United States) Infinity Lab LC/MDS-system with ESI-source. The column temperature was 40 °C. The samples (1 mg/mL) were dissolved in methanol containing 2 mmol/L ammonium acetate and 2 µL of this solution were injected. The method was a run with a gradient from 90% 2 mmol/L ammonium acetate in H_2O to 100% acetonitrile within 10 min and consecutively a flush with 100% acetonitrile for 10 min with a flowrate of 0.5 mL/min. The UV absorption was detected from 190 – 600 nm using a diode array detector (DAD) and purity was determined at 220 – 600 nm, if not mentioned otherwise. Positive and negative total ion scans were recorded from 100 – 1000 *m/z* or more if necessary. Otherwise, LC-MS analyses (method B) were obtained with an HPLC HP1100 (Agilent) using an EC50/2 Nucleodur C18 Gravity 3 µm column (Macherey-Nagel) coupled with an API2000 mass spectrometer (ABSciex) with ESI-source. The column temperature was 25 °C. The samples (1 mg/mL) were dissolved in methanol, containing 2 mmol/L ammonium acetate, and 8 µL were injected. The method was a run (0.3 mL/min) with a gradient from 90% 2 mmol/L ammonium acetate in H_2O to 100% 2 mmol/L ammonium acetate in methanol within 10 min and a subsequent flush with 100% of 2 mmol/L ammonium acetate in methanol. The UV absorption was detected from 190 – 900 nm using a diode array detector (DAD) and purity was determined at 220 – 400 nm, if not stated otherwise. Positive and negative total ion scans were recorded from 150 – 800 *m/z*. HRMS spectra (method C) were measured on an LTQ Orbitrap XL (Thermo Fisher Scientific, Bremen, Germany) with HPLC autosampler. Samples were dissolved in acetonitrile. Positive or negative full scans were observed from 200 – 1000 *m/z* or more if necessary. Alternatively, HRMS data (method D) were obtained using an HPLC Dionex Ultimate 3000 (Thermo Scientific) using flow injection mode coupled with a micrOTOF-Q mass spectrometer (Bruker) with ESI-source. The method run (0.3 mL/min) was an isocratic flush of 0.1% acetic acid in acetonitrile. Positive or negative full scans were recorded from 50 – 1000 *m/z* (or more if necessary). Sodium acetate was used as calibrant.

4.2. Preparation of the Titrant 1

*(S)-Benzyl 1-(1-methylhydrazinyl)-1-oxo-3-phenylpropan-2-ylcarbamate (**6**)* [9-12]. Cbz-Phe-OH (6.00 g, 1.00 eq, 20.05 mmol) was dissolved in 60 mL anhyd THF. *N*-methyl morpholine (2.03 g, 1.00 eq, 20.05 mmol, 2.23 mL) and isobutyl chloroformate (2,74 g, 1.00 eq, 20.05 mmol, 2.61 mL) were added to the stirring solution at 0 °C. Subsequently, methylhydrazine (4.62 g, 5.00 eq, 100.23 mmol, 5.25 mL) was added and the mixture was stirred for 18 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between EtOAc and H_2O (each 150 mL). The phases were separated and the aqueous layer was extracted with EtOAc (2 × 75 mL). The combined organic layers were washed with brine (100 mL) dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by column chromatography (CH_2Cl_2/MeOH 39:1) to yield the described product as a white solid. (3.61 g, 55% yield); R_f = 0.36 (CH_2Cl_2/MeOH 29:1); mp 122– 124 °C, lit.[9] mp 120 – 121 °C. ^{1}H NMR (600 MHz, DMSO-d_6) δ 2.64 (dd, *J* = 13.6, 10.3 Hz, 1H, CHC<u>H</u>$_2$Ph), 3.04 (s, 4H, CHC<u>H</u>$_2$Ph, NCH_3), 4.85 (s, 2H, NH_2), 4.92 (d, *J* = 3.0 Hz, 2H, CH_2OCO), 5.17 (td, *J* = 9.5, 9.0, 3.2 Hz, 1H, C<u>H</u>CH_2Ph), 7.06 – 7.37 (m, 11H, CH_{arom}, OCONH) ppm. ^{13}C NMR (151 MHz, DMSO-d_6) δ 36.72 (CH<u>C</u>H_2Ph), 37.63 (NCH_3), 52.85 (<u>C</u>HCH_2Ph), 64.98 (<u>C</u>H_2OCO), 126.09 (C-4), 127.43 (C-2', C-6'), 127.60 (C-4'), 127.97 (C-2, C-6), 128.23 (C-3', C-5'), 129.05 (C-3, C-5), 137.14 (C-1'), 138.84 (C-1), 155.75 (OCON), 172.93 (<u>C</u>ONCH_3) ppm. LC-MS (ESI), t_R = 10.26 min, 99% purity (method B), *m/z* calcd for $[C_{18}H_{21}N_3O_3+H]^+$, 328.2, found, 328.0.

*Benzyl (S,E)-(1-(2-benzylidene-1-methylhydrazinyl)-1-oxo-3-phenylpropan-2-yl)carbamate (**7**)*[11,12]. Benzaldehyde (2.20 g, 2.00 eq, 20.77 mmol, 2.10 mL) was added to a stirring solution of semicarbazide **6** (3.40 g, 1.00 eq, 10.39 mmol) and acetic acid (1.25g, 2.00 eq, 20.77 mmol, 1.19 mL) in 40 mL EtOH/H_2O (4:1) and the mixture was stirred for 18 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between EtOAc and H_2O (each 100 mL). The phases were separated and the aqueous layer was extracted with EtOAc (2 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by column chromatography (CH_2Cl_2/MeOH 100:0 → 98.5:1.5) to obtain the desired product as a white solid. (3.79 g, 88% yield); R_f = 0.17 (petroleum ether/EtOAc 4:1); mp 131 – 133 °C , lit.[11] mp 128 – 130 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 2.82 (dd, *J* = 13.7, 10.1 Hz, 1H, CHC<u>H</u>$_2$Ph), 3.01 (dd, *J* = 13.7, 4.0 Hz, 1H, CHC<u>H</u>$_2$Ph), 3.34 (s, 3H, NCH_3), 4.96 (s, 2H, CH_2OCO), 5.50 (ddd, *J* = 10.3, 8.8, 4.0 Hz, 1H, C<u>H</u>CH_2Ph), 7.15 – 7.23 (m, 1H), 7.23 – 7.37 (m, 9H), 7.40 – 7.52 (m, 3H), 7.69 (d, *J* = 8.8 Hz, 1H, OCONH), 7.74 – 7.80 (m, 2H), 8.01 (s, 1H, N=CH) ppm. ^{13}C NMR (126 MHz, DMSO-d_6) δ 28.20 (NCH_3), 36.86 (CH<u>C</u>H_2Ph), 53.47 (<u>C</u>HCH_2Ph), 65.18 (<u>C</u>H_2OCO), 126.30 (C-4), 126.96 (C-3'', C-5''), 127.47 (C-2', C-6'), 127.63 (C-4'), 128.07 (C-2, C-6), 128.22 (C-3', C-5'), 128.73 (C-2'', C-6''), 128.95 (C-3, C-5), 129.70 (C-4''), 134.61 (C-1''), 136.99 (C-1'), 138.11 (C-1), 140.93 (N=CH), 155.91 (OCON), 172.68 ($CONCH_3$) ppm. LC-MS (ESI), t_R = 12.03 min, > 99% purity (method B), *m/z* calcd for $[C_{25}H_{25}N_3O_3+H]^+$, 416.2, found, 416.1.

*tert-Butyl ((S)-1-(((S)-1-(2-benzyl-1-methylhydrazinyl)-1-oxo-3-phenylpropan-2-yl)amino)-3,3-dimethyl-1-oxobutan-2-yl)carbamate (**8**).* A Schlenk flask, containing 10% palladium on activated carbon (200 mg), was evacuated, refilled with argon and evacuated again (three reruns). Subsequently, a solution of Cbz-protected amine **7** (2.00 g, 1.00 eq, 4.81 mmol) in 16 mL anhyd THF and 4 mL anhyd ethanol was added and the reaction vessel was set under hydrogen atmosphere. The corresponding mixture was stirred for 20 h at ambient temperature. The suspension was filtered through a plug of celite, washed with EtOAc (3 × 50 mL) and all volatiles were removed

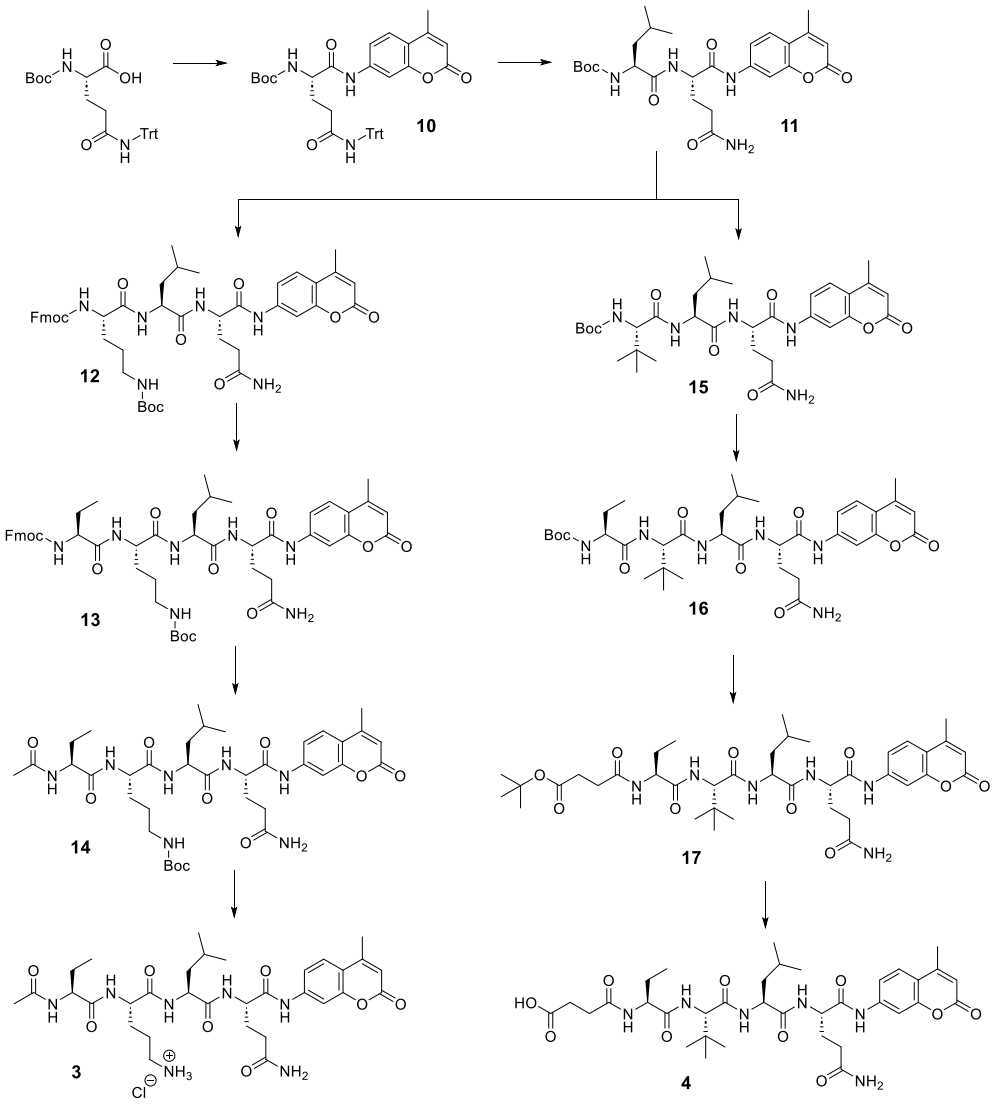
under reduced pressure. HATU (2.01 g, 1.10 eq, 5.29 mmol) was added to a stirring mixture of Boc-Tle-OH (1.22 g, 1.10 eq, 5.29 mmol) and DIPEA (3.11 g, 5.00 eq, 24.07 mmol, 4.09 mL) in 10 mL anhyd DMF at 0 °C under argon atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. The deprotected amine, dissolved in 10 mL anhyd DMF, was added dropwise to the solution at 0 °C and the reaction was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure. The residue was partitioned between EtOAc and H_2O (each 100 mL) and the phases were separated. The aqueous layer was extracted with EtOAc (2 × 75 mL) and the combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and removed under reduced pressure. The crude material was purified by column chromatography (CH_2Cl_2/MeOH 100:0 → 98.5:1.5) to yield the described product as a white, voluminous solid. (1.83 g, 77% yield); R_f = 0.45 (CH_2Cl_2/MeOH 19:1); mp 74 – 78 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.77 (s, 9H, CHC(C<u>H</u>$_3$)$_3$), 1.38 (s, 9H, OC(CH$_3$)$_3$), 2.68 – 2.76 (m, 1H, CHC<u>H</u>$_2$Ph), 2.87 – 2.94 (m, 1H, CHC<u>H</u>$_2$Ph), 3.04 (s, 3H, NCH$_3$), 3.78 – 3.95 (m, 3H, NHC<u>H</u>$_2$Ph, C<u>H</u>C(CH$_3$)$_3$), 5.25 – 5.35 (m, 1H, N<u>H</u>CH$_2$Ph), 5.61 (td, *J* = 8.9, 4.5 Hz, 1H, C<u>H</u>CH$_2$Ph), 6.44 (d, *J* = 9.9 Hz, 1H, N<u>H</u>CHC(CH$_3$)$_3$), 7.06 – 7.48 (m, 10H), 7.78 (d, *J* = 8.6 Hz, 1H, N<u>H</u>CHCH$_2$Ph) ppm. ^{13}C NMR (126 MHz, DMSO-d6) δ 26.53 (CHC(<u>C</u>H$_3$)$_3$), 28.12 (OC(<u>C</u>H$_3$)$_3$), 32.10 (NCH$_3$), 34.04 (CH<u>C</u>(CH$_3$)$_3$), 37.27 (CH<u>C</u>H$_2$Ph), 49.59 (NHCH$_2$), 52.38 (<u>C</u>HCH$_2$Ph), 62.09 (<u>C</u>HC(CH$_3$)$_3$), 78.00 (O<u>C</u>(CH$_3$)$_3$), 126.04 (C-4), 127.26 (C-4''), 127.83 (C-2, C-6), 128.17 (C-2'', C-6''), 128.89 (C-3'', C-5''), 129.01 (C-3, C-5), 137.25 (C-1''), 137.95 (C-1), 155.02 (OCON), 169.61 (<u>C</u>ONHCHCH$_2$Ph), 172.86 (<u>C</u>ONCH$_3$) ppm. LC-MS (ESI), t_R = 12.01 min, 94% purity (method B, DAD 200 – 400 nm), *m/z* calcd for $[C_{28}H_{40}N_4O_4+H]^+$, 497.3, found, 497.4.

*Benzyl ((S)-1-(((S)-1-(((S)-1-(2-benzyl-1-methylhydrazinyl)-1-oxo-3-phenylpropan-2-yl)amino)-3,3-dimethyl-1-oxobutan-2-yl)amino)-1-oxobutan-2-yl)carbamate (**9**).* Boc-protected amine **8** (1.80 mg, 1.00 eq, 3.62 mmol) was stirred for 2 h in 4 mL anhyd CH_2Cl_2 and 4 mL 4 N HCl in 1,4-dioxane at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with diethylether (3 × 15 mL) and the residue was dried under high vacuum. EDC·HCl (764 mg, 1.10 eq, 3.99 mmol) and HOBt·H_2O (611 mg, 1.10 eq, 3.99 mmol) were added to a stirring mixture of Cbz-Abu-OH (946 mg, 1.10 eq, 3.99 mmol) and DIPEA (3.28 g, 7.00 eq, 25.34 mmol, 4.31 mL) in 10 mL anhyd CH_2Cl_2 at 0 °C under argon atmosphere and the mixture was stirred for 15 min at 0 °C. The deprotected amine, dissolved in 10 mL anhyd CH_2Cl_2, was added dropwise at 0 °C and the resulting mixture was stirred for 16 h at ambient temperature. The solution was partitioned between CH_2Cl_2 and H_2O (each 100 mL). The phases were separated and the aqueous layer was extracted with CH_2Cl_2 (2 × 75 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 97:3) to obtain a white solid. (1.34 g, 60%); R_f = 0.24 (CH_2Cl_2/MeOH 19:1); mp 91 – 94 °C.^{1}H NMR (500 MHz, DMSO-d_6) δ 0.67 – 0.98 (m, 12H, C(C<u>H</u>$_3$)$_3$, CH$_2$C<u>H</u>$_3$), 1.42 – 1.54 (m, 1H, C<u>H</u>$_2$CH$_3$), 1.54 – 1.65 (m, 1H, C<u>H</u>$_2$CH$_3$), 2.75 (dd, *J* = 13.9, 8.7 Hz, 1H, CHC<u>H</u>$_2$Ph), 2.91 (dd, *J* = 13.9, 4.7 Hz, 1H, CHC<u>H</u>$_2$Ph), 3.03 (s, 3H, NCH$_3$), 3.79 – 3.92 (m, 2H, NHC<u>H</u>$_2$Ph), 3.92 – 4.07 (m, 1H, C<u>H</u>CH$_2$CH$_3$), 4.26 (d, *J* = 9.5 Hz, 1H, C<u>H</u>C(CH$_3$)$_3$), 5.02 (s, 2H, CH$_2$OCO), 5.27 (t, *J* = 6.4 Hz, 1H, N<u>H</u>CH$_2$Ph), 5.56 (td, *J* = 8.5, 4.8 Hz, 1H, C<u>H</u>CH$_2$Ph), 7.07 – 7.21 (m, 5H), 7.20 – 7.39 (m, 10H), 7.43 (d, *J* = 8.2 Hz, 1H, CONH), 7.47 (d, *J* = 9.7 Hz, 1H, CONH), 7.89 (d, *J* = 8.3 Hz, 1H, CONH) ppm. ^{13}C NMR (126 MHz, DMSO-d_6) δ 10.33 (CH$_2$<u>C</u>H$_3$), 25.10 (<u>C</u>H$_2$CH$_3$), 26.54 (C(<u>C</u>H$_3$)$_3$), 32.14 (NCH$_3$), 34.52 (<u>C</u>(CH$_3$)$_3$), 37.01 (CH<u>C</u>H$_2$Ph), 49.50 (NH<u>C</u>H$_2$Ph), 52.35 (<u>C</u>HCH$_2$Ph), 56.12, 59.28 (<u>C</u>HCH$_2$CH$_3$, <u>C</u>HC(CH$_3$)$_3$), 65.27 (<u>C</u>H$_2$OCO), 125.99 (C-4), 127.23 (C-4''), 127.58 (C-2', C-6'), 127.65 (C-4'), 127.82 (C-2, C-6), 128.14 (C-2'', C-6''), 128.23 (C-3', C-5'), 128.87 (C-3'', C-5'' and C-3, C-

5), 137.00 (C-1'), 137.25 (C-1''), 137.93 (C-1), 155.87 (OCON), 169.36 (CONH), 171.16 (CONH), 172.66 (CONCH$_3$) ppm. LC-MS (ESI), t_R = 11.95 min, 98% purity (method B), *m/z* calcd for $[C_{35}H_{45}N_5O_5+H]^+$, 616.3, found, 616.8.

*Benzyl ((S)-1-(((S)-1-(((S)-1-(2-benzyl-2-cyano-1-methylhydrazinyl)-1-oxo-3-phenylpropan-2-yl)amino)-3,3-dimethyl-1-oxobutan-2-yl)amino)-1-oxobutan-2-yl)carbamate (**1**).* Cyanogen bromide (1.34 g, 6.00 eq, 12.67 mmol) was added to a solution of acylhydrazine derivative **9** (1.30 g, 1.00 eq, 2.11 mmol) and sodium acetate (1.04 g, 6.00 eq, 12.67 mmol) in 20 mL anhyd methanol. The reaction mixture was stirred for 18 h at ambient temperature. All volatiles were removed under reduced pressure. The residue was diluted with 100 mL H_2O and the aqueous layer was adjusted to pH 1-2 with 10% aq. $KHSO_4$. The aqueous layer was extracted with EtOAc (3 × 70 mL) and the combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 97:3) to achieve the desired product as a white solid. (772 mg, 57% yield); R_f = 0.27 (CH_2Cl_2/MeOH 19:1); mp 96 – 99 °C. ^{1}H NMR (500 MHz, DMSO-d_6, 70 °C) δ 0.85 (t, *J* = 7.4 Hz, 3H, CH_2CH_3), 0.91 (s, 9H, $C(CH_3)_3$), 1.51 – 1.62 (m, 1H, CH_2CH_3), 1.62 – 1.74 (m, 1H, CH_2CH_3), 2.88 – 3.00 (m, 2H, $CHCH_2Ph$), 3.06 (s, 3H, NCH_3), 4.03 (td, *J* = 8.4, 5.6 Hz, 1H, $CHCH_2CH_3$), 4.17 – 4.83 (m, 3H, $CHC(CH_3)_3$, NCH_2Ph), 4.88 – 5.39 (m, 3H, CH_2OCO, $CHCH_2Ph$), 6.87 – 7.71 (m, 17H), 8.32 (s, 1H, CONH) ppm. ^{13}C NMR (126 MHz, DMSO-d_6, 70 °C) δ 9.76 (CH_2CH_3), 24.83 (CH_2CH_3), 26.20 ($C(CH_3)_3$), 34.17 ($C(CH_3)_3$), 36.54 ($CHCH_2Ph$), 49.87, 56.06, 56.85, 58.99 (NCH_2Ph, 3 × $NHCCO$), 65.17 (CH_2OCO), 112.11 (C≡N), 126.23, 127.15, 127.29, 127.88, 128.34, 128.60, 129.52 (9 × CH_{arom}), 132.25 (C_{arom}), 136.36 (C_{arom}), 136.72 (C_{arom}), 155.53 (OCON), 169.83 (CONH), 170.91 (2 × CONH) ppm (NCH_3 signal is not visible). LC-MS (ESI), t_R = 11.81 min, 95% purity (method A, DAD 220 – 600 nm), *m/z* calcd for $[C_{36}H_{44}N_6O_5 + H]^+$, 641.3, found, 641.5. HRMS m/z calcd for $[C_{36}H_{44}N_6O_5+Na]^+$, 663.3265, found, 663.3270 (method D).

4.3. Preparation of Fluorogenic Substrates



Scheme S1. Synthesis of AMC substrates.

tert-Butyl (S)-(1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxo-5-(tritylamino)pentan-2-yl)carbamate ***(10)****.* Phosphoryl chloride (494 mg, 1.05 eq, 3.22 mmol, 301 µL) was added dropwise to a stirring solution of Boc-Gln(Trt)-OH (1.50 g, 1.00 eq, 3.07 mmol) and 7-amino-4-methyl-2*H*-chromen-2-one (538 mg, 1.00 eq, 3.07 mmol) in 30 mL anhyd pyridine at -10 °C under nitrogen atmosphere and the corresponding mixture was stirred for 1 h at 0 °C. The reaction was quenched carefully by the addition of water (100 mL). EtOAc (100 mL) was added, the phases were separated and the aqueous layer was extracted with EtOAc (3 × 50 mL). The combined organic layers were washed with sat. $NaHCO_{3aq}$ solution (100 mL), 10% HSO_{4aq} solution (100 mL) and with brine (100 mL), dried over Na_2SO_4, filtered and all volatiles were removed under reduced pressure. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 98:2) to obtain the desired product as a white solid. (710 mg, 36% yield); R_f = 0.32 (CH_2Cl_2/MeOH 19:1); mp 175 – 180 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 1.39 (s, 9H, C(CH_3)$_3$), 1.71 – 1.82 (m, 1H, CHCH_2C$\underline{H}_2$), 1.82 – 1.93 (m, 1H, CHCH_2C$\underline{H}_2$), 2.27 – 2.37 (m, 1H, CHC$\underline{H}_2$CH_2), 2.37 – 2.47 (m, 4H, CHC$\underline{H}_2$CH_2, C-4'-CH_3), 4.06 (dt, *J* = 12.9, 6.2 Hz, 1H, C$\underline{H}$CH_2CH_2), 6.26 (d, *J* = 1.1 Hz, 1H, 3'-H), 7.06 (d, *J* = 7.4

Hz, 1H, OCONH), 7.13 – 7.22 (m, 9H, CH_{arom}), 7.23 – 7.30 (m, 6H, CH_{arom}), 7.49 (d, *J* = 8.6 Hz, 1H, 6'-H), 7.72 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.77 (d, *J* = 1.9 Hz, 1H, 8'-H), 8.56 (s, 1H, CONHTrt), 10.37 (s, 1H, CHCONH) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 17.87 (C-4'-CH_3), 27.49 ($CHCH_2CH_2$), 28.13 ($C(CH_3)_3$), 32.76 ($CHCH_2CH_2$), 54.97 ($CHCH_2CH_2$), 69.17 (CONH$C(Ph_3)$), 78.23 ($C(CH_3)_3$), 105.70 (C-8'), 112.22 (C-3'), 114.99 (C-4a'), 115.22 (C-6'), 125.83 (C-5'), 126.25 (Trt-C_{para}), 127.38 (Trt-C_{ortho}), 128.44 (Trt-C_{metha}), 142.20 (C-7'), 144.81 (Trt-C_{ipso}), 152.97 (C-4'), 153.59 (C-8a'), 155.39 (OCON), 159.91 (C-2'), 171.15 ,(CHCONH) 171.66 (CONHTrt) ppm. LC-MS (ESI), t_R = 8.56 min, > 99% purity (method A), *m/z* calcd for $[C_{39}H_{39}N_3O_6 + H]^+$, 646.3, found, 646.5.

*tert-Butyl ((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)carbamate (**11**).* A solution of Boc-protected amine **10** (690 mg, 1.00 eq, 1.07 mmol) and triisopropylsilane (508 mg, 3.00 eq, 3.21 mmol, 657 µL) in 5 mL anhyd CH_2Cl_2 and 5 mL TFA was stirred for 2 h at ambient temperature. All volatiles were removed in vacuum, coevaporated with CH_2Cl_2 (3 × 15 mL) and dried under high vacuum. HATU (488 mg, 1.20 eq, 1.28 mmol) was added to a stirring solution of Boc-*L*-Leu-OH (297 mg, 1.20 eq, 1.28 mmol) in 10 mL anhyd DMF and DIPEA (691 mg, 5.00 eq, 5.34 mmol, 931 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, the deprotected amine, dissolved in 8 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between CH_2Cl_2 and H_2O (each 75 mL). The aqueous layer was adjusted to pH 10–12 with saturated aqueous $NaHCO_3$ solution, the phases were separated and the aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 94:6) to yield the titled compound as a white solid. (433 mg, 78% yield); R_f = 0.44 (CH_2Cl_2/MeOH 9:1); mp 193 – 196 °C, lit.[1] mp 170 – 172 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.86 (d, *J* = 6.6 Hz, 3H, $CH(CH_3)_2$), 0.88 (d, *J* = 6.7 Hz, 3H, $CH(CH_3)_2$), 1.38 (s, 9H, $C(CH_3)_3$), 1.41 – 1.48 (m, 2H, $CH_2CH(CH_3)_2$), 1.55 – 1.71 (m, 1H, $CH(CH_3)_2$), 1.81 – 1.92 (m, 1H, $CHCH_2CH_2$), 1.92 – 2.01 (m, 1H, $CHCH_2CH_2$), 2.06 – 2.23 (m, 2H, $CHCH_2CH_2$), 2.40 (s, 3H, C-4'-CH_3), 3.86 – 4.06 (m, 1H, NHCHCO), 4.32 – 4.47 (m, 1H, NHCHCO), 6.27 (d, *J* = 1.5 Hz, 1H, 3'-H), 6.77 (s, 1H, $CONH_2$), 6.93 (d, *J* = 8.1 Hz, 1H, OCONH), 7.27 (s, 1H, $CONH_2$), 7.49 (dd, *J* = 8.7, 1.9 Hz, 1H, 6'-H), 7.73 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.76 (d, *J* = 2.0 Hz, 1H, 8'-H), 8.07 (d, *J* = 7.5 Hz, 1H, CONHCHCH$_2$CH$_2$), 10.43 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 17.87 (C-4'-CH_3), 21.43 ($CH(CH_3)_2$), 22.95 ($CH(CH_3)_2$), 24.20 ($CH(CH_3)_2$), 27.64 ($CHCH_2CH_2$), 28.12 ($C(CH_3)_3$), 31.16 ($CHCH_2CH_2$), 40.60 ($CH_2CH(CH_3)_2$) 52.83 (NHCHCO), 53.14 (NHCHCO), 78.09 ($C(CH_3)_3$), 105.76 (C-8'), 112.30 (C-3'), 115.12 (C-4a'), 115.25 (C-6'), 125.85 (C-5'), 142.00 (C-7'), 152.96 (C-4'), 153.57 (C-8a'), 155.38 (OCON), 159.89 (C-2'), 170.91 (CONH), 172.66 (CONH), 173.33 ($CONH_2$) ppm. LC-MS (ESI), t_R = 5.88 min, > 99% purity (method A), *m/z* calcd for $[C_{26}H_{36}N_4O_7 + H]^+$, 517.3, found, 517.5.

(9H-Fluoren-9-yl)methyl tert-butyl ((S)-5-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-5-oxopentane-1,4-diyl)dicarbamate ***(12)***. Boc protected amine derivative **11** (370 mg, 1.00 eq, 716 µmol) was stirred for 2 h in 4 mL anhyd CH_2Cl_2 and 4 mL TFA at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH_2Cl_2 (3 × 15 mL) and the residue was dried under high vacuum. HATU (327 mg, 1.20 eq, 859 µmol) was added to a stirring solution of (*S*)-2-((((9*H*-fluoren-9-yl)methoxy)carbonyl)amino)-5-((*tert*-butoxycarbonyl)amino)pentanoic acid (391 mg, 1.20 eq, 859 µmol) in 10 mL anhyd DMF and DIPEA (463 mg, 5.00 eq, 3.58 mmol, 624 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, the deprotected amine, dissolved in 10 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 3 days at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between EtOAc and H_2O (each 75 mL). The aqueous layer was adjusted to pH 10–12 with saturated aqueous $NaHCO_3$ solution, the phases were separated and the aqueous layer was extracted with EtOAc (3 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 99:1 → 94:6) to obtain the titled compound as a white solid. (315 mg, 52% yield); R_f = 0.33 (CH_2Cl_2/MeOH 9:1); mp 185 – 187 °C. ^{1}H NMR (600 MHz, DMSO-d_6) δ 0.84 (d, *J* = 6.5 Hz, 3H, $CH_2CH(C\underline{H}_3)_2$), 0.87 (d, *J* = 6.6 Hz, 3H, $CH_2CH(C\underline{H}_3)_2$), 1.32 – 1.56 (m, 14H, $C(C\underline{H}_3)_3$, $C\underline{H}_2CH(CH_3)_2$, $CHCH_2C\underline{H}_2CH_2NH$, $CHC\underline{H}_2CH_2CH_2NH$), 1.57 – 1.69 (m, 2H, $CH_2C\underline{H}(CH_3)_2$, $CHC\underline{H}_2CH_2CH_2NH$), 1.83 – 1.92 (m, 1H, $CHCH_2C\underline{H}_2CO$), 1.93 – 2.02 (m, 1H, $CHCH_2C\underline{H}_2CO$), 2.10 – 2.23 (m, 2H, $CHC\underline{H}_2CH_2CO$), 2.40 (d, *J* = 1.3 Hz, 3H, C-4'-CH_3), 2.84 – 2.99 (m, 2H, $CHCH_2CH_2C\underline{H}_2NH$), 4.00 (td, *J* = 8.9, 4.8 Hz, 1H, $NHC\underline{H}CO$), 4.14 – 4.40 (m, 5H, 2 × $NHC\underline{H}CO$, $C\underline{H}C\underline{H}_2OCONH$), 6.26 (d, *J* = 1.6 Hz, 1H, 3'-H), 6.68 – 6.85 (m, 2H, $CONH_2$, $CH_2CH_2CH_2N\underline{H}$), 7.27 (s, 1H, $CONH_2$), 7.29 – 7.35 (m, 2H), 7.38 – 7.44 (m, 2H), 7.47 – 7.53 (m, 2H, CONH, 6'-H), 7.68 – 7.74 (m, 3H), 7.76 (d, *J* = 2.0 Hz, 1H, 8'-H), 7.83 – 7.93 (m, 3H), 8.18 (d, *J* = 7.1 Hz, 1H, CONH), 10.40 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (151 MHz, DMSO-d_6) δ 17.93 (C-4'-$\underline{C}H_3$), 21.54 ($CH(\underline{C}H_3)_2$), 23.06 ($CH(\underline{C}H_3)_2$), 24.07 ($\underline{C}H(CH_3)_2$), 26.17 ($CHCH_2\underline{C}H_2CH_2NH$), 27.41 ($CH\underline{C}H_2CH_2CO$), 28.24 ($C(\underline{C}H_3)_3$), 29.18 ($CH\underline{C}H_2CH_2CH_2NH$), 31.27 ($CHCH_2\underline{C}H_2CO$), 40.06 ($CHCH_2CH_2\underline{C}H_2NH$), 40.69 ($\underline{C}H_2CH(CH_3)_2$), 46.65 ($\underline{C}HCH_2OCON$), 50.93 ($NH\underline{C}HCO$), 53.39 ($NH\underline{C}HCO$), 54.41 ($NH\underline{C}HCO$), 65.61 ($CH\underline{C}H_2OCON$), 77.38 ($\underline{C}(CH_3)_3$), 105.74 (C-8'), 112.32 (C-3'), 115.10 (C-4a'), 115.28 (C-6'), 120.06, 125.26, 125.89 (C-5'), 127.04, 127.58, 140.67, 142.09 (C-7'), 143.70, 143.87, 153.02 (C-4'), 153.61 (C-8a'), 155.57 (OCON), 155.97 (OCON), 159.96 (C-2'), 170.91 (CONH), 171.88 (CONH), 172.12 (CONH), 173.37 ($CONH_2$) ppm. LC-MS (ESI), t_R = 7.66 min, 95% purity (method A, DAD 285 – 287 nm), *m/z* calcd for frag. $[C_{41}H_{48}N_6O_8 + H]^+$, 753.4, found, 753.6.

*tert-Butyl ((S)-4-((S)-2-((((9H-fluoren-9-yl)methoxy)carbonyl)amino)butanamido)-5-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-5-oxopentyl)carbamate (**13**).* Fmoc-protected amine **12** (300 mg, 1.00 eq, 352 µmol) was stirred in 5 mL anhyd DMF and 5 mL morpholine for 18 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was dried under high vacuum. HATU (174 mg, 1.30 eq, 457 µmol) was added to a stirring solution of Fmoc-Abu-OH (149 mg, 1.30 eq, 457 µmol) in 10 mL anhyd DMF and DIPEA (227 mg, 5.00 eq, 1.76 mmol, 306 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, deprotected amine, dissolved in 12 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure and dried under high vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 99:1 → 93:7) to obtain the titled product as a pale brown solid. (316 mg, 91% yield); R_f = 0.38 (CH_2Cl_2/MeOH 9:1); mp 192 – 194 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.73 – 0.95 (m, 9H, $CH_2C\underline{H}_3$, $CH(C\underline{H}_3)_2$), 1.29 – 1.73 (m, 18H, $C(C\underline{H}_3)_3$, $C\underline{H}_2C\underline{H}(CH_3)_2$, $CHC\underline{H}_2C\underline{H}_2CH_2NH$, $C\underline{H}_2CH_3$), 1.82 – 2.02 (m, 2H, $CHCH_2C\underline{H}_2CO$), 2.09 – 2.23 (m, 2H, $CHC\underline{H}_2CH_2CO$), 2.40 (s, 3H, C-4'-CH_3), 2.80 – 2.95 (m, 2H, $CHCH_2CH_2C\underline{H}_2NH$), 3.95 (td, *J* = 8.1, 5.1 Hz, 1H, $NHC\underline{H}CO$), 4.14 – 4.42 (m, 6H, 3 × $NHC\underline{H}CO$, $C\underline{H}C\underline{H}_2OONH$), 6.27 (d, *J* = 1.0 Hz, 1H, 3'-H), 6.71 (t, *J* = 5.8 Hz, 1H, $CH_2CH_2CH_2N\underline{H}$), 6.79 (s, 1H, $CONH_2$), 7.28 (s, 1H, $CONH_2$), 7.32 (t, *J* = 7.4 Hz, 2H), 7.41 (t, *J* = 7.5 Hz, 2H), 7.46 (d, *J* = 8.0 Hz, 1H, CONH), 7.50 (dd, *J* = 8.5, 2.2 Hz, 1H, 6'-H), 7.68 – 7.76 (m, 3H), 7.76 (d, *J* = 1.8 Hz, 1H, 8'-H), 7.82 – 7.93 (m, 3H), 7.99 (d, *J* = 7.8 Hz, 1H, CONH), 8.15 (d, *J* = 7.0 Hz, 1H, CONH), 10.38 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 10.21 ($CH_2\underline{C}H_3$), 17.87 (C-4'-$\underline{C}H_3$), 21.44 ($CH(\underline{C}H_3)_2$), 22.99 ($CH(\underline{C}H_3)_2$), 24.02 ($\underline{C}H(CH_3)_2$), 25.19 ($\underline{C}H_2CH_3$), 25.97 ($CHCH_2\underline{C}H_2CH_2NH$), 27.40 ($CH\underline{C}H_2CH_2CO$), 28.17 ($C(\underline{C}H_3)_3$), 29.07 ($CH\underline{C}H_2CH_2CH_2NH$), 31.24 ($CHCH_2\underline{C}H_2CO$), 40.60 ($\underline{C}H_2CH(CH_3)_2$), 46.63 ($\underline{C}HCH_2OCON$), 50.99 ($NH\underline{C}HCO$), 52.37 ($NH\underline{C}HCO$), 53.35 ($NH\underline{C}HCO$), 55.94 ($NH\underline{C}HCO$), 65.59 ($CH\underline{C}H_2OCON$), 77.31 ($\underline{C}(CH_3)_3$), 105.75 (C-8'), 112.29 (C-3'), 115.09 (C-4a'), 115.26 (C-6'), 119.99, 125.22, 125.82 (C-5'), 126.97, 127.54, 140.64, 142.04 (C-7'), 143.71, 143.81, 152.95 (C-4'), 153.58 (C-8a'), 155.51 (OCON), 155.93 (OCON), 159.89 (C-2'), 170.84 (CONH), 171.49 (CONH), 171.80 (CONH), 171.95 (CONH), 173.36 ($CONH_2$) ppm (one signal is obscured by the DMSO signal). LC-MS (ESI), t_R = 6.62 min, 91% purity (method A), *m/z* calcd for $[C_{50}H_{63}N_7O_{11}+H]^+$, 938.5, found, 938.7.

*tert-Butyl ((S)-4-((S)-2-acetamidobutanamido)-5-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-5-oxopentyl)carbamate (**14**).* Fmoc-protected amine **13** (300 mg, 1.00 eq, 320 µmol) was stirred in 5 mL anhyd DMF and 5 mL morpholine for 18 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was dried under high vacuum. The deprotected amine was dissolved in 10 mL anhyd CH_2Cl_2 and 10 mL anhyd DMF and the mixture was cooled down to 0 °C and set under nitrogen atmosphere. DIPEA (207 mg, 5.00 eq, 1.60 mmol, 279 µL) and subsequently acetyl chloride (30 mg, 1.20 eq, 384 µmol, 27.4 µL) were added and the resulting mixture was stirred for 3 days at ambient temperature. All volatiles were removed under reduced pressure and the residue was dried under high vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 99:1 → 93:7) to yield the desired product as a white solid. (172 mg, 71% yield); R_f = 0.31 (CH_2Cl_2/MeOH 9:1); mp 204 – 215 °C. ^{1}H NMR (600 MHz, DMSO-d_6) δ 0.81 – 0.86 (m, 6H, $CH_2C\underline{H}_3$, $CH(C\underline{H}_3)_2$), 0.88 (d, *J* = 6.6 Hz, 3H, $CH(C\underline{H}_3)_2$), 1.31 – 1.56 (m, 16H, $C(C\underline{H}_3)_3$, $C\underline{H}_2CH_3$, $C\underline{H}_2CH(CH_3)_2$, $CHCH_2C\underline{H}_2CH_2NH$, $CHC\underline{H}_2CH_2CH_2NH$), 1.57 – 1.70 (m, 2H, $CH_2C\underline{H}(CH_3)_2$, $CHC\underline{H}_2CH_2CH_2NH$), 1.82 – 1.93 (m, 4H, CH_3CO, $CHCH_2C\underline{H}_2CO$), 1.92 – 2.01 (m, 1H, $CHCH_2C\underline{H}_2CO$), 2.10 – 2.23

(m, 2H, CHC<u>H</u>$_2$CH$_2$CO), 2.40 (s, 3H, C-4'-CH$_3$), 2.84 – 2.94 (m, 2H, CHCH$_2$CH$_2$C<u>H</u>$_2$NH), 4.13 – 4.24 (m, 2H, 2 × NHC<u>H</u>CO), 4.26 – 4.37 (m, 2H, 2 × NHC<u>H</u>CO), 6.27 (d, *J* = 1.5 Hz, 1H, 3'-H), 6.74 (t, *J* = 5.9 Hz, 1H, CH$_2$CH$_2$CH$_2$N<u>H</u>), 6.78 (s, 1H, CONH$_2$), 7.29 (s, 1H, CONH$_2$), 7.52 (dd, *J* = 8.8, 2.0 Hz, 1H, 6'-H), 7.73 (d, *J* = 8.6 Hz, 1H, 5'-H), 7.77 (d, *J* = 2.0 Hz, 1H, 8'-H), 7.82 (d, *J* = 7.9 Hz, 1H, CONH), 7.97 (d, *J* = 7.7 Hz, 1H, CONH), 8.00 (d, *J* = 7.9 Hz, 1H, CONH), 8.13 (d, *J* = 7.3 Hz, 1H, CONH), 10.41 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (151 MHz, DMSO-*d*$_6$) δ 10.31 (CH$_2$<u>C</u>H$_3$), 18.11 (C-4'-<u>C</u>H$_3$), 21.61 (CH(<u>C</u>H$_3$)$_2$), 22.63 (<u>C</u>H$_3$CO), 23.27 (CH(<u>C</u>H$_3$)$_2$), 24.22 (<u>C</u>H(CH$_3$)$_2$), 25.37 (<u>C</u>H$_2$CH$_3$), 26.26 (CHCH$_2$<u>C</u>H$_2$CH$_2$NH), 27.61 (CH<u>C</u>H$_2$CH$_2$CO), 28.41 (C(CH$_3$)$_3$), 29.10 (CH<u>C</u>H$_2$CH$_2$CH$_2$NH), 31.47 (CHCH$_2$<u>C</u>H$_2$CO), 38.62 (CHCH$_2$CH$_2$<u>C</u>H$_2$NH), 40.74 (<u>C</u>H$_2$CH(CH$_3$)$_2$), 51.19 (NH<u>C</u>HCO), 52.66 (NH<u>C</u>HCO), 53.59 (NH<u>C</u>HCO), 54.17 (NH<u>C</u>HCO), 77.54 (<u>C</u>(CH$_3$)$_3$), 105.92 (C-8'), 112.48 (C-3'), 115.28 (C-4a'), 115.47 (C-6'), 126.06 (C-5'), 142.30 (C-7'), 153.22 (C-4'), 153.78 (C-8a'), 155.74 (OCON), 160.14 (C-2'), 169.59 (CONH), 171.11 (CONH), 171.73 (CONH), 172.01 (CONH), 172.21 (CONH), 173.57 (CONH$_2$) ppm. LC-MS (ESI), t_R = 5.54 min, 98% purity (method A), *m/z* calcd for [C$_{37}$H$_{55}$N$_7$O$_{10}$ + H]$^+$, 758.4, found, 758.7.

*(S)-4-((S)-2-Acetamidobutanamido)-5-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-5-oxopentan-1-aminium chloride (**3**).* Boc protected amine derivative **14** (150 mg, 1.00 eq, 198 µmol) was stirred for 2 h in 4 mL anhyd CH$_2$Cl$_2$ and 4 mL 4 N HCl in dioxane at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH$_2$Cl$_2$ (3 × 15 mL) and the residue was dried under high vacuum to obtain the titled product as a pale brown solid. (137 mg, quantitavive yield); R_f = 0.30 (CH$_2$Cl$_2$/7 N NH$_3$ in MeOH 3:1); mp 190 – 196 °C (decomposition). ^{1}H NMR (600 MHz, DMSO-*d*$_6$) δ 0.81 – 0.86 (m, 6H, CH$_2$C<u>H</u>$_3$, CH(C<u>H</u>$_3$)$_2$), 0.88 (d, *J* = 6.6 Hz, 3H, CH(C<u>H</u>$_3$)$_2$), 1.45 – 1.70 (m, 8H), 1.73 – 1.80 (m, 1H) (C<u>H</u>$_2$CH$_3$, C<u>H</u>$_2$C<u>H</u>(CH$_3$)$_2$, CHC<u>H</u>$_2$C<u>H</u>$_2$CH$_2$NH), 1.83 – 1.94 (m, 4H, CH$_3$CO, CHCH$_2$C<u>H</u>$_2$CO), 1.94 – 2.02 (m, 1H, CHCH$_2$C<u>H</u>$_2$CO), 2.11 – 2.23 (m, 2H, CHC<u>H</u>$_2$CH$_2$CO), 2.40 (d, *J* = 1.3 Hz, 3H, C-4'-CH$_3$), 2.74 – 2.83 (m, 2H, CHCH$_2$CH$_2$C<u>H</u>$_2$NH), 4.17 (td, *J* = 7.9, 5.6 Hz, 1H, NHC<u>H</u>CO), 4.27 – 4.34 (m, 2H, 2 × NHC<u>H</u>CO), 4.37 (td, *J* = 8.0, 5.5 Hz, 1H, NHC<u>H</u>CO), 6.27 (d, *J* = 1.6 Hz, 1H, 3'-H), 6.80 (s, 1H, CONH$_2$), 7.34 (s, 1H, CONH$_2$), 7.54 (dd, *J* = 8.7, 2.1 Hz, 1H, 6'-H), 7.73 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.79 (d, *J* = 2.1 Hz, 1H, 8'-H), 7.85 – 7.97 (m, 3H, NH$_3$), 7.98 – 8.06 (m, 2H, 2 × CONH), 8.11 (d, *J* = 8.1 Hz, 1H, CONH), 8.19 (d, *J* = 7.3 Hz, 1H, CONH), 10.58 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (151 MHz, DMSO-*d*$_6$) δ 10.17 (CH$_2$<u>C</u>H$_3$), 17.96 (C-4'-<u>C</u>H$_3$, obscured), 21.45 (CH(<u>C</u>H$_3$)$_2$), 22.47 (<u>C</u>H$_3$CO), 23.09 (CH(<u>C</u>H$_3$)$_2$), 23.46 (CHCH$_2$<u>C</u>H$_2$CH$_2$NH), 24.07 (<u>C</u>H(CH$_3$)$_2$), 25.26 (<u>C</u>H$_2$CH$_3$), 27.54 (CH<u>C</u>H$_2$CH$_2$CO), 28.38 (CH<u>C</u>H$_2$CH$_2$CH$_2$NH), 31.33 (CHCH$_2$<u>C</u>H$_2$CO), 38.25 (CHCH$_2$CH$_2$<u>C</u>H$_2$NH), 40.61 (<u>C</u>H$_2$CH(CH$_3$)$_2$), 51.12 (NH<u>C</u>HCO), 51.69 (NH<u>C</u>HCO), 53.41 (NH<u>C</u>HCO), 54.02 (NH<u>C</u>HCO), 105.74 (C-8'), 112.30 (C-3'), 115.09 (C-4a'), 115.31 (C-6'), 125.89 (C-5'), 142.18 (C-7'), 153.08 (C-4'), 153.60 (C-8a'), 159.99 (C-2'), 169.45 (CONH), 170.97 (CONH), 171.24 (CONH), 171.91 (CONH), 172.02 (CONH), 173.50 (CONH$_2$) ppm. LC-MS (ESI), t_R = 4.56 min, 95% purity (method A), *m/z* calcd for [C$_{32}$H$_{47}$N$_7$O$_8$ + H]$^+$, 658.4, found, 658.5. HRMS m/z calcd for [C$_{32}$H$_{47}$N$_7$O$_8$+ H]$^+$, 658.3559, found, 658.3562 (method C).

*tert-Butyl ((S)-1-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-3,3-dimethyl-1-oxobutan-2-yl)carbamate (**15**).* Boc protected amine derivative **11** (410 mg, 1.00 eq, 794 µmol) was stirred for 2 h in 5 mL anhyd CH_2Cl_2 and 5 mL TFA at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH_2Cl_2 (3 × 20 mL) and the residue was dried under high vacuum. HATU (362 mg, 1.20 eq, 952 µmol) was added to a stirring solution of Boc-*L*-Tle-OH (220 mg, 1.20 eq, 952 µmol) in 10 mL anhyd DMF and DIPEA (513 mg, 5.00 eq, 3.97 mmol, 691 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, the deprotected amine, dissolved in 8 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between CH_2Cl_2 and H_2O (each 75 mL). The aqueous layer was adjusted to pH 10–12 with saturated aqueous $NaHCO_3$ solution, the phases were separated and the aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 99:1 → 94:6) to yield the titled compound as a white solid. (470 mg, 94% yield); R_f = 0.35 (CH_2Cl_2/MeOH 9:1); mp 197 – 201 °C, lit.[1] mp 182 – 183 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.84 (d, *J* = 6.5 Hz, 3H, CH(C$\underline{H}_3$)$_2$), 0.86 – 0.93 (m, 12H, CH(C$\underline{H}_3$)$_2$, CHC(C$\underline{H}_3$)$_3$), 1.38 (s, 9H, OC(C$\underline{H}_3$)$_3$), 1.42 – 1.52 (m, 2H, C$\underline{H}_2$CH(CH$_3$)$_2$), 1.58 – 1.72 (m, 1H, C$\underline{H}$(CH$_3$)$_2$), 1.79 – 2.01 (m, 2H, CHCH$_2$C$\underline{H}_2$), 2.07 – 2.24 (m, 2H, CHC$\underline{H}_2$CH$_2$), 2.40 (s, 3H, C-4'-CH$_3$), 3.87 (d, *J* = 9.6 Hz, 1H, C$\underline{H}$C(CH$_3$)$_3$), 4.28 – 4.44 (m, 2H, 2 × NHC$\underline{H}$CO), 6.27 (d, *J* = 1.4 Hz, 1H, 3'-H), 6.55 (d, *J* = 9.7 Hz, 1H, N$\underline{H}$CHC(CH$_3$)$_3$), 6.77 (s, 1H, CONH$_2$), 7.26 (s, 1H, CONH$_2$), 7.48 (dd, *J* = 8.7, 2.0 Hz, 1H, 6'-H), 7.72 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.76 (d, *J* = 1.9 Hz, 1H, 8'-H), 7.88 (d, *J* = 7.8 Hz, 1H, CONH), 8.18 (d, *J* = 6.8 Hz, 1H, CONH), 10.45 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 17.88 (C-4'-$\underline{C}$H$_3$), 21.37 (CH(C$\underline{H}_3$)$_2$), 23.01 (CH(C$\underline{H}_3$)$_2$), 23.97 (C$\underline{H}$(CH$_3$)$_2$), 26.60 (CHC(C$\underline{H}_3$)$_3$), 27.45 (CH$\underline{C}$H$_2$CH$_2$), 28.07 (OC(C$\underline{H}_3$)$_3$), 31.24 (CHCH$_2$C$\underline{H}_2$), 33.89 (CH$\underline{C}$(CH$_3$)$_3$), 40.67 (C$\underline{H}_2$CH(CH$_3$)$_2$), 50.77 (NH$\underline{C}$HCO), 53.37 (NH$\underline{C}$HCO), 62.08 (C$\underline{H}$C(CH$_3$)$_3$), 78.10 (O$\underline{C}$(CH$_3$)$_3$), 105.71 (C-8'), 112.26 (C-3'), 115.06 (C-4a'), 115.23 (C-6'), 125.84 (C-5'), 142.08 (C-7'), 152.97 (C-4'), 153.58 (C-8a'), 155.27 (OCON), 159.90 (C-2'), 170.20 (CONH), 170.84 (CONH), 172.01 (CONH), 173.28 (CONH$_2$) ppm. LC-MS (ESI), t_R = 6.47 min, 99% purity (method A) , *m/z* calcd for [$C_{32}H_{47}N_5O_8$+ H]$^+$, 630.3, found, 630.5.

*tert-Butyl ((S)-1-(((S)-1-(((S)-1-(((S)-5-amino-1-((4-methyl-2-oxo-2H-chromen-7-yl)amino)-1,5-dioxopentan-2-yl)amino)-4-methyl-1-oxopentan-2-yl)amino)-3,3-dimethyl-1-oxobutan-2-yl)amino)-1-oxobutan-2-yl)carbamate (**16**).* Boc protected amine derivative **15** (450 mg, 1.00 eq, 715 µmol) was stirred for 2 h in 4 mL anhyd CH_2Cl_2 and 4 mL TFA at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH_2Cl_2 (3 × 20 mL) and the residue was dried under high vacuum. HATU (326 mg, 1.20 eq, 857 µmol) was added to a stirring solution of Boc-Abu-OH (174 mg, 1.20 eq, 857 µmol) in 10 mL anhyd DMF and DIPEA (462 mg, 5.00 eq, 3.57 mmol, 622 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, the deprotected amine, dissolved in 8 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between CH_2Cl_2 and H_2O (each 75 mL). The aqueous layer was adjusted to pH 10–12 with saturated aqueous $NaHCO_3$ solution, the phases were separated and the aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 94:6) to

yield the titled compound as a white solid. (489 mg, 96% yield); R_f = 0.38 (CH_2Cl_2/MeOH 9:1); mp 178 – 183 °C, lit.[1] mp 168 – 170 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.79 – 0.86 (m, 6H, $CH_2C\underline{H}_3$, $CH(C\underline{H}_3)_2$), 0.85 – 0.93 (m, 12H, $CH(C\underline{H}_3)_2$, $CHC(C\underline{H}_3)_3$), 1.37 (s, 9H, $OC(C\underline{H}_3)_3$), 1.41 – 1.54 (m, 3H, $C\underline{H}_2CH(CH_3)_2$, $C\underline{H}_2CH_3$), 1.53 – 1.67 (m, 2H, $C\underline{H}(CH_3)_2$, $C\underline{H}_2CH_3$), 1.81 – 1.99 (m, 2H, $CHCH_2C\underline{H}_2$), 2.07 – 2.23 (m, 2H, $CHC\underline{H}_2CH_2$), 2.40 (s, 3H, C-4'-CH_3), 3.87 (td, *J* = 8.3, 5.6 Hz, 1H, $C\underline{H}CH_2CH_3$), 4.27 (d, *J* = 9.3 Hz, 1H, $C\underline{H}C(CH_3)_2$), 4.30 – 4.40 (m, 2H, 2 × NHC<u>H</u>CO), 6.27 (d, *J* = 1.5 Hz, 1H, 3'-H), 6.78 (s, 1H, $CONH_2$), 7.02 (d, *J* = 8.1 Hz, 1H, OCONH), 7.26 (s, 1H, $CONH_2$), 7.43 – 7.51 (m, 2H, 6'-H, CONH), 7.73 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.76 (d, *J* = 1.9 Hz, 1H, 8'-H), 8.07 (d, *J* = 8.0 Hz, 1H, CONH), 8.12 (d, *J* = 7.0 Hz, 1H, CONH), 10.46 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 10.25 ($CH_2\underline{C}H_3$), 17.88 (C-4'-$\underline{C}H_3$), 21.40 ($CH(\underline{C}H_3)_2$), 22.92 ($CH(\underline{C}H_3)_2$), 24.06 ($\underline{C}H(CH_3)_2$), 24.83 ($\underline{C}H_2CH_3$), 26.50 ($CHC(\underline{C}H_3)_3$), 27.52 ($CH\underline{C}H_2CH_2$), 28.10 ($OC(\underline{C}H_3)_3$), 31.23 ($CHCH_2\underline{C}H_2$), 34.47 ($CH\underline{C}(CH_3)_3$), 40.53 ($\underline{C}H_2CH(CH_3)_2$), 50.78 (NH<u>C</u>HCO), 53.37 (NH<u>C</u>HCO), 55.92 (NH<u>C</u>HCO), 59.27 (NH<u>C</u>HCO), 78.01 ($O\underline{C}(CH_3)_3$), 105.71 (C-8'), 112.27 (C-3'), 115.07 (C-4a'), 115.23 (C-6'), 125.84 (C-5'), 142.07 (C-7'), 152.97 (C-4'), 153.58 (C-8a'), 155.27 (OCON), 159.90 (C-2'), 169.81 (CONH), 170.82 (CONH), 171.63 (CONH), 171.82 (CONH), 173.27 ($CONH_2$) ppm. LC-MS (ESI), t_R = 6.60 min, 99% purity (method A), *m/z* calcd for $[C_{36}H_{54}N_6O_9$+ H]$^+$, 715.4, found, 715.6.

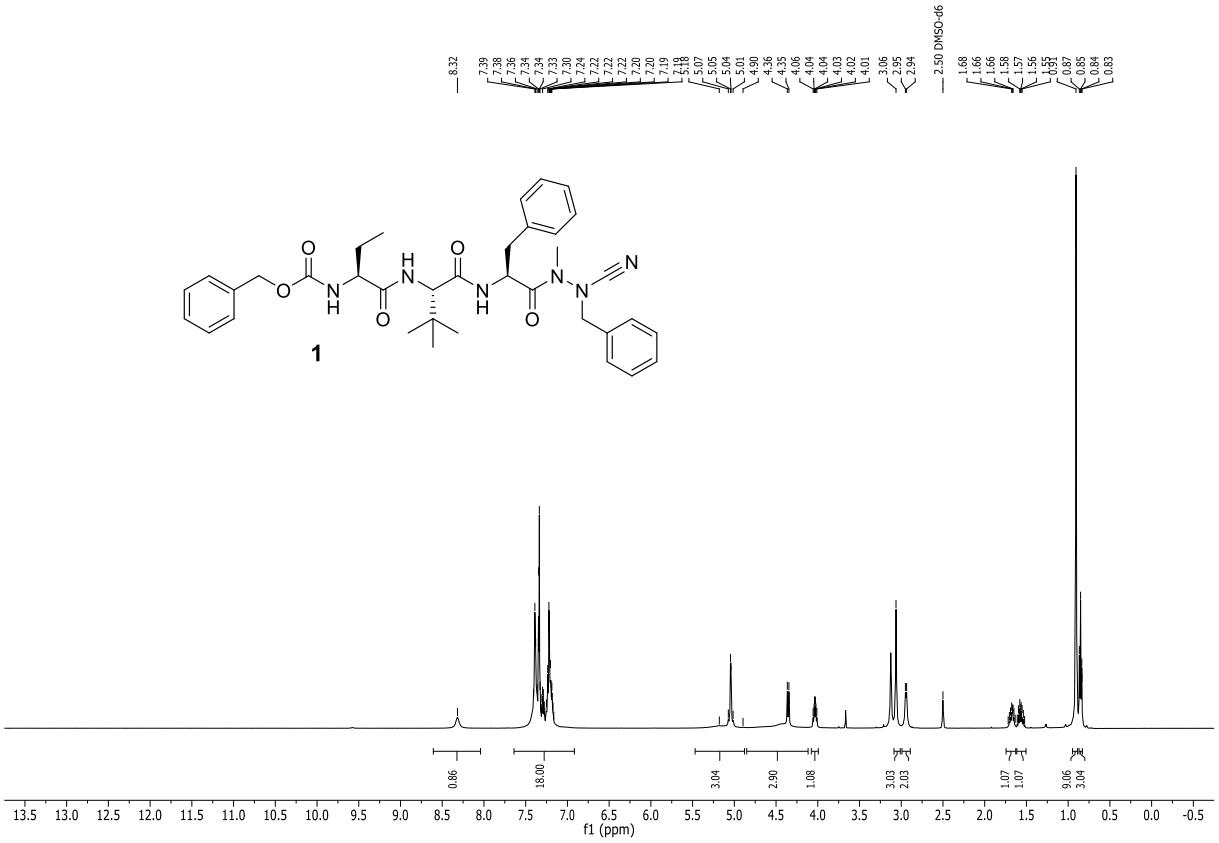
*tert-Butyl (6S,9S,12S,15S)-18-amino-9-(tert-butyl)-6-ethyl-12-isobutyl-15-((4-methyl-2-oxo-2H-chromen-7-yl)carbamoyl)-4,7,10,13,18-pentaoxo-5,8,11,14-tetraazaoctadecanoate (**17**)*. Boc protected amine derivative **16** (320 mg, 1.00 eq, 448 µmol) was stirred for 2 h in 5 mL anhyd CH_2Cl_2 and 5 mL TFA at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH_2Cl_2 (3 × 20 mL) and the residue was dried under high vacuum. HATU (204 mg, 1.20 eq, 537 µmol) was added to a stirring solution of 4-(*tert*-butoxy)-4-oxobutanoic acid (94 mg, 1.20 eq, 537 µmol) in 10 mL anhyd DMF and DIPEA (289 mg, 5.00 eq, 2.24 mmol, 390 µL) at 0 °C under nitrogen atmosphere and the corresponding mixture was stirred for 15 min at 0 °C. Subsequently, the deprotected amine, dissolved in 8 mL anhyd DMF, was added dropwise at 0 °C and the reaction mixture was stirred for 16 h at ambient temperature. All volatiles were removed under reduced pressure and the residue was partitioned between CH_2Cl_2 and H_2O (each 75 mL). The aqueous layer was adjusted to pH 10–12 with saturated aqueous $NaHCO_3$ solution, the phases were separated and the aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. The crude material was purified by flash chromatography (CH_2Cl_2/MeOH 100:0 → 92:8) to obtain the titled compound as a colorless solid. (310 mg, 90% yield); R_f = 0.33 (CH_2Cl_2/MeOH 9:1); mp 152 – 156 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.78 – 0.84 (m, 6H, $CH_2C\underline{H}_3$, $CH(C\underline{H}_3)_2$), 0.88 (d, *J* = 6.6 Hz, 3H, $CH(C\underline{H}_3)_2$), 0.90 (s, 9H, $CHC(C\underline{H}_3)_3$), 1.37 (s, 9H, $OC(CH_3)_3$), 1.42 – 1.55 (m, 3H, $C\underline{H}_2CH(CH_3)_2$, $C\underline{H}_2CH_3$), 1.55 – 1.68 (m, 2H, $C\underline{H}(CH_3)_2$, $C\underline{H}_2CH_3$), 1.81 – 1.99 (m, 2H, $CHCH_2C\underline{H}_2$), 2.07 – 2.22 (m, 2H, $CHC\underline{H}_2CH_2$), 2.29 – 2.43 (m, 7H, C-4'-CH_3, $OCOC\underline{H}_2C\underline{H}_2CO$), 4.23 (d, *J* = 9.2 Hz, 1H, $C\underline{H}C(CH_3)_3$), 4.28 (td, *J* = 7.9, 5.7 Hz, 1H, NHC<u>H</u>CO), 4.31 – 4.39 (m, 2H, 2 × NHC<u>H</u>CO), 6.27 (d, *J* = 1.5 Hz, 1H, 3'-H), 6.78 (s, 1H, $CON\underline{H}_2$), 7.26 (s, 1H, $CON\underline{H}_2$), 7.48 (dd, *J* = 8.7, 2.0 Hz, 1H, 6'-H), 7.62 (d, *J* = 9.3 Hz, 1H, CONH), 7.73 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.76 (d, *J* = 2.0 Hz, 1H, 8'-H), 7.97 (d, *J* = 7.9 Hz, 1H, CONH), 8.01 (d, *J* = 8.0 Hz, 1H, CONH), 8.08 (d, *J* = 7.2 Hz, 1H, CONH), 10.44 (s, 1H, CONHPh) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 9.98 ($CH_2\underline{C}H_3$), 17.88 (C-4'-$\underline{C}H_3$), 21.30 ($CH(\underline{C}H_3)_2$), 22.99 ($CH(\underline{C}H_3)_2$), 24.06 ($\underline{C}H(CH_3)_2$), 25.01 ($\underline{C}H_2CH_3$), 26.56 ($CHC(\underline{C}H_3)_3$), 27.52 ($CH\underline{C}H_2CH_2$), 27.67 ($OC(\underline{C}H_3)_3$), 29.94 ($OCOCH_2\underline{C}H_2CON$), 30.27 ($OCO\underline{C}H_2CH_2CON$), 31.22 ($CHCH_2\underline{C}H_2$), 34.05 ($CH\underline{C}(CH_3)_3$), 40.48 ($\underline{C}H_2CH(CH_3)_2$), 50.84 (NH<u>C</u>HCO), 53.36

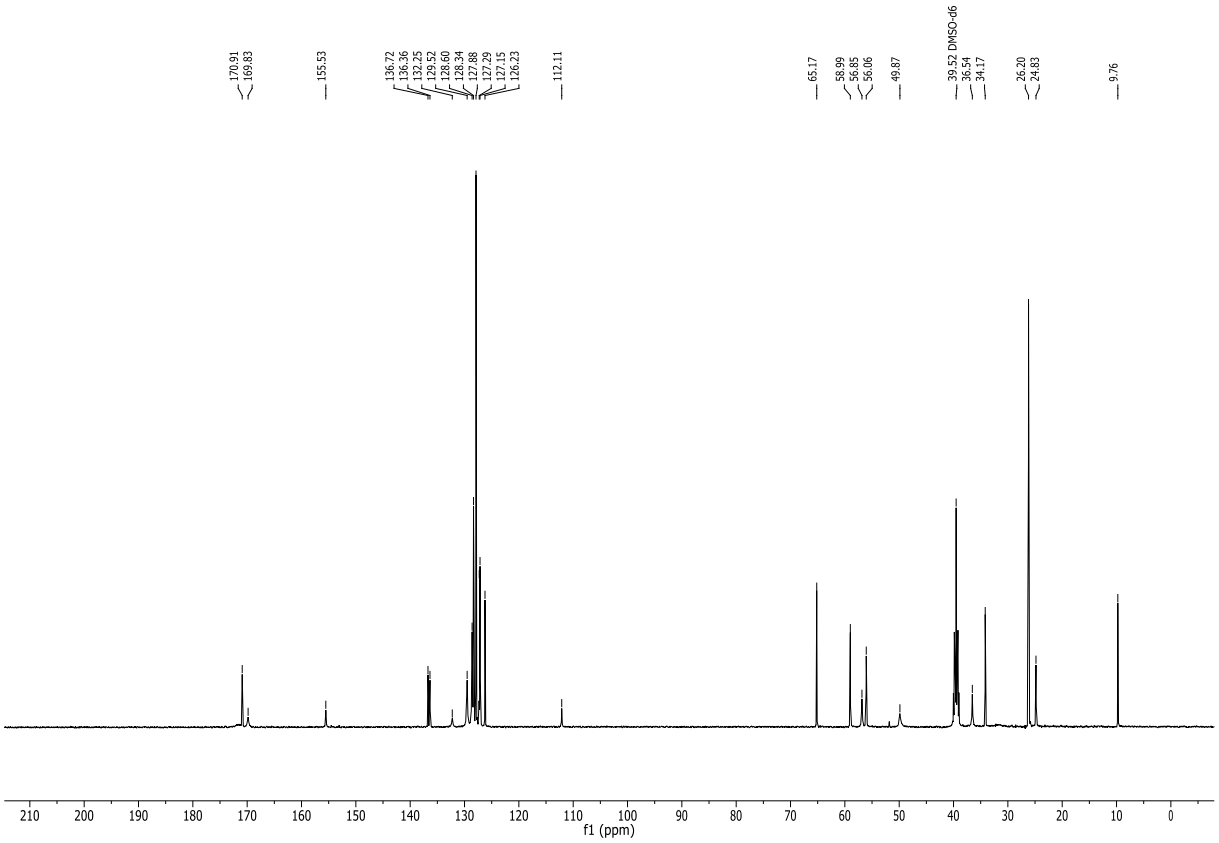
(NHCHCO), 53.84 (NHCHCO), 59.72 (NHCHCO), 79.44 (OC(CH$_3$)$_3$), 105.72 (C-8'), 112.27 (C-3'), 115.07 (C-4a'), 115.24 (C-6'), 125.84 (C-5'), 142.05 (C-7'), 152.97 (C-4'), 153.58 (C-8a'), 159.90 (C-2'), 169.89 (CONH), 170.81 (CONH), 171.38, 171.40 (CONH, COO), 171.86 (CONH), 173.27 (CONH$_2$) ppm (one signal is not visible). LC-MS (ESI), t_R = 6.65 min, 99% purity (method A), *m/z* calcd for [$C_{39}H_{58}N_6O_{10}$+ H]$^+$, 771.4, found, 771.6.

*(6S,9S,12S,15S)-18-Amino-9-(tert-butyl)-6-ethyl-12-isobutyl-15-((4-methyl-2-oxo-2H-chromen-7-yl)carbamoyl)-4,7,10,13,18-pentaoxo-5,8,11,14-tetraazaoctadecanoic acid (**4**).* *tert*-Butyl protected carboxylic acid derivative **17** (290 mg, 1.00 eq, 376 µmol) was stirred for 2 h in 4 mL anhyd CH_2Cl_2 and 4 mL 4 N HCl in dioxane at ambient temperature. All volatiles were removed under reduced pressure, coevaporated with CH_2Cl_2 (3 × 20 mL) and the residue was dried under high vacuum. The crude material was purified by reversed phase flash chromatography (H_2O/ACN 95:5 → 50:50) to obtain the titled product as a white solid. (190 mg, 71% yield); R_f = 0.32 (CH_2Cl_2/MeOH 9:1 + 1% AcOH); mp 165 – 170 °C. ^{1}H NMR (500 MHz, DMSO-d_6) δ 0.79 – 0.85 (m, 6H, CH$_2$CH$_3$, CH(CH$_3$)$_2$), 0.88 (d, *J* = 6.6 Hz, 3H, CH(CH$_3$)$_2$), 0.90 (s, 9H, CHC(CH$_3$)$_3$), 1.40 – 1.55 (m, 3H, CH$_2$CH(CH$_3$)$_2$, CH$_2$CH$_3$), 1.55 – 1.69 (m, 2H, CH(CH$_3$)$_2$, CH$_2$CH$_3$), 1.81 – 1.99 (m, 2H, CHCH$_2$CH$_2$), 2.07 – 2.23 (m, 2H, CHCH$_2$CH$_2$), 2.28 – 2.47 (m, 7H, C-4'-CH$_3$, HOOCCH$_2$CH$_2$CO), 4.22 (d, *J* = 9.3 Hz, 1H, CHC(CH$_3$)$_3$), 4.27 (td, *J* = 7.9, 5.7 Hz, 1H, NHCHCO), 4.30 – 4.39 (m, 2H, 2 × NHCHCO), 6.27 (d, *J* = 1.5 Hz, 1H, 3'-H), 6.77 (s, 1H, CONH$_2$), 7.26 (s, 1H, CONH$_2$), 7.48 (dd, *J* = 8.6, 2.1 Hz, 1H, 6'-H), 7.60 (d, *J* = 9.2 Hz, 1H, CONH), 7.73 (d, *J* = 8.7 Hz, 1H, 5'-H), 7.76 (d, *J* = 2.0 Hz, 1H, 8'-H), 7.96 – 8.04 (m, 2H, 2 × CONH), 8.08 (d, *J* = 7.1 Hz, 1H, CONH), 10.43 (s, 1H, CONHPh), 12.04 (s, 1H, COOH) ppm. ^{13}C NMR {^{1}H} (126 MHz, DMSO-d_6) δ 9.99 (CH$_2$CH$_3$), 17.88 (C-4'-CH$_3$), 21.31 (CH(CH$_3$)$_2$), 22.98 (CH(CH$_3$)$_2$), 24.06 (CH(CH$_3$)$_2$), 24.94 (CH$_2$CH$_3$), 26.56 (CHC(CH$_3$)$_3$), 27.51 (CHCH$_2$CH$_2$), 29.16 (HOOCCH$_2$CH$_2$CO), 29.89 (HOOCCH$_2$CH$_2$CO), 31.23 (CHCH$_2$CH$_2$), 34.07 (CHC(CH$_3$)$_3$), 40.46 (CH$_2$CH(CH$_3$)$_2$), 50.85 (NHCHCO), 53.37 (NHCHCO), 53.87 (NHCHCO), 59.73 (NHCHCO), 105.72 (C-8'), 112.28 (C-3'), 115.08 (C-4a'), 115.24 (C-6'), 125.84 (C-5'), 142.06 (C-7'), 152.98 (C-4'), 153.58 (C-8a'), 159.90 (C-2'), 169.90 (CONH), 170.83 (CONH), 171.05 (CONH), 171.37 (CONH), 171.87 (CONH), 173.29 (CONH$_2$), 173.64 (COOH) ppm. LC-MS (ESI), t_R = 4.30 min, >99% purity (method A), *m/z* calcd for [$C_{35}H_{50}N_6O_{10}$+ H]$^+$, 715.4, found, 715.6. HRMS m/z calcd for [$C_{35}H_{50}N_6O_{10}$+ H]$^+$, 715.3661, found, 715.3666 (method C).

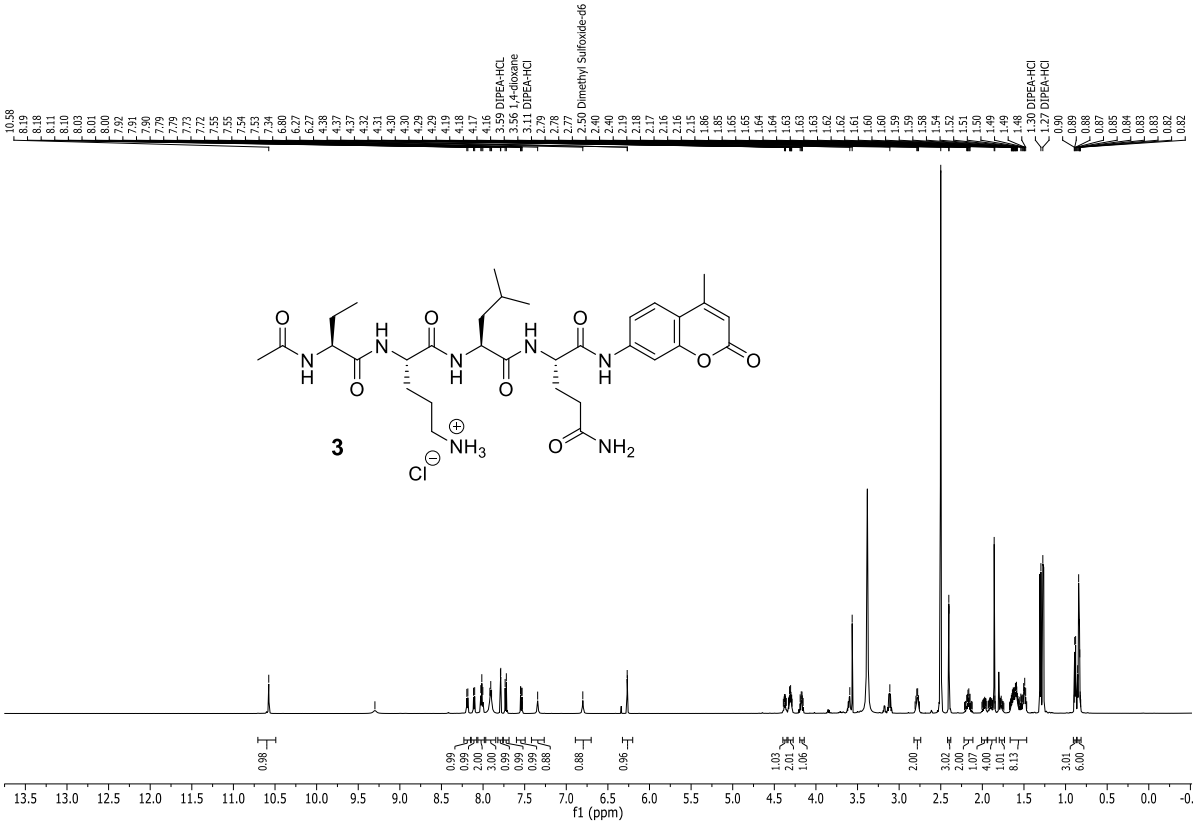
4.4. ^{1}H and ^{13}C NMR Spectra

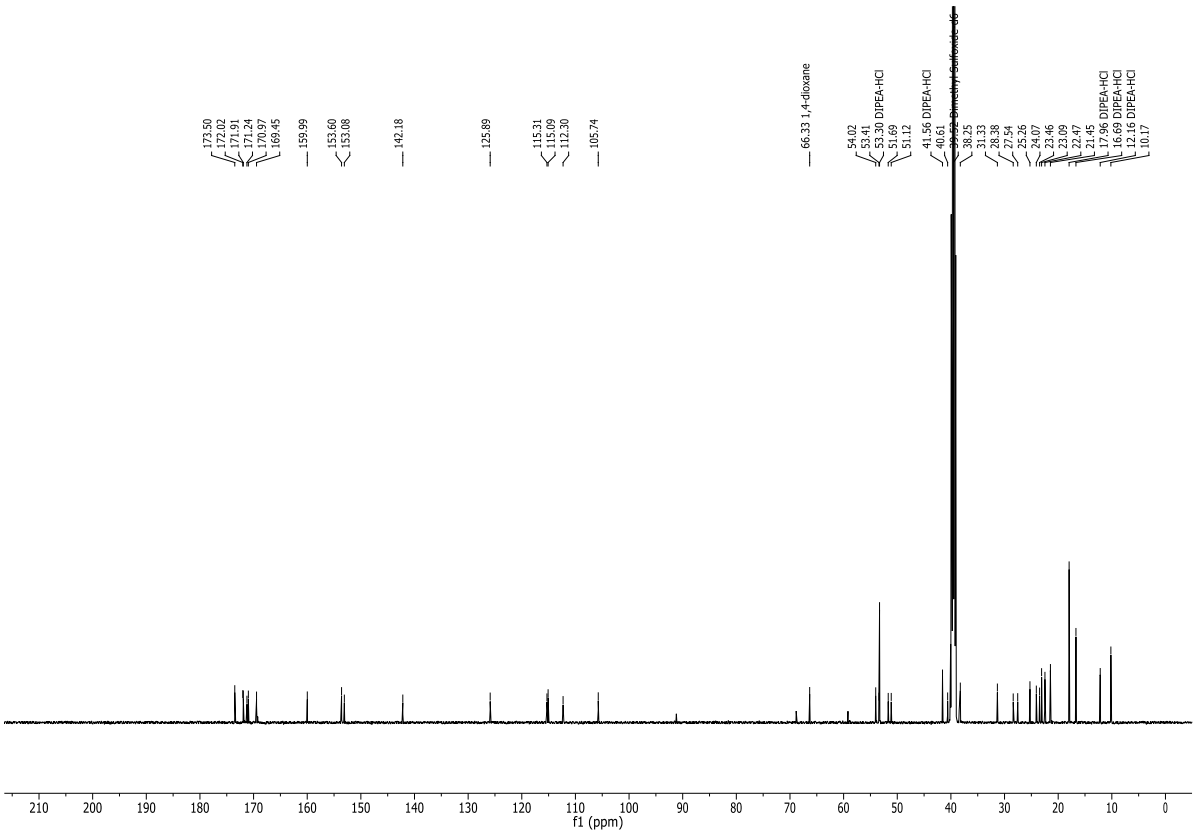
4.4.1. ^{1}H NMR and ^{13}C NMR Spectra (at 70 °C) of Compound 1



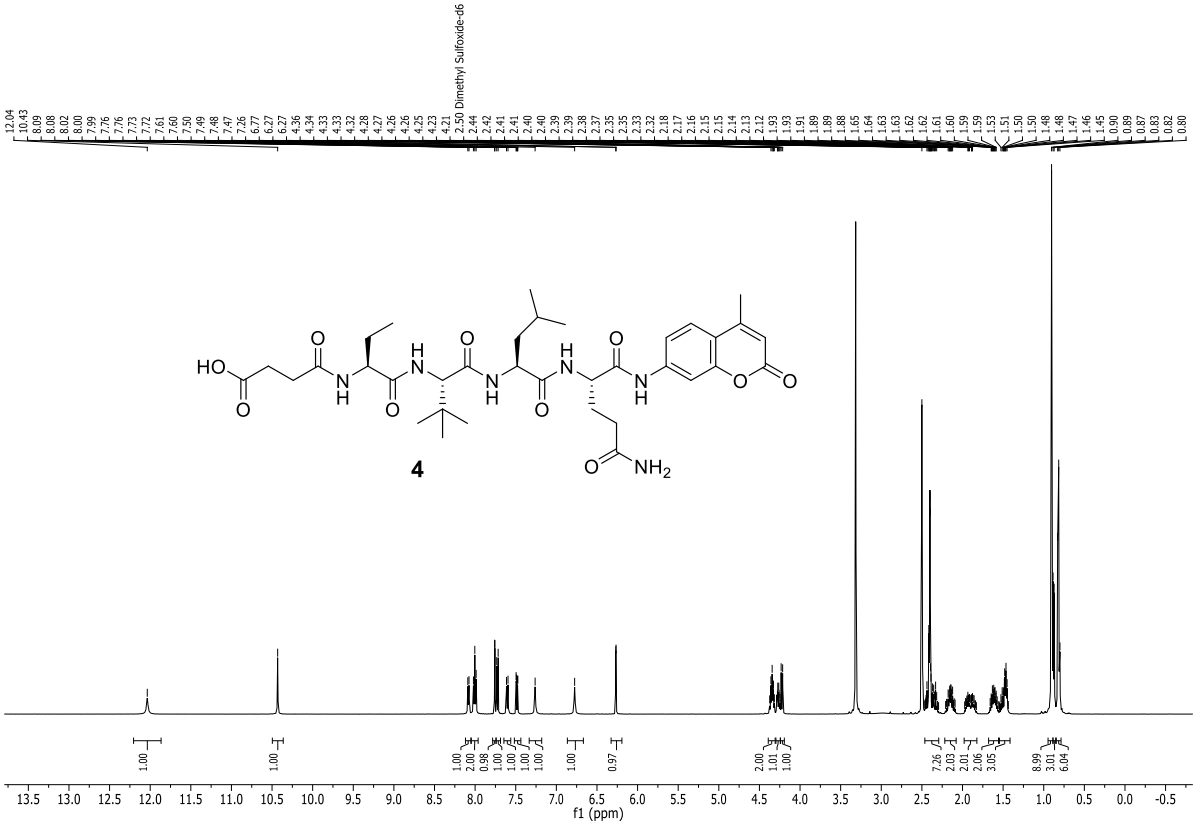


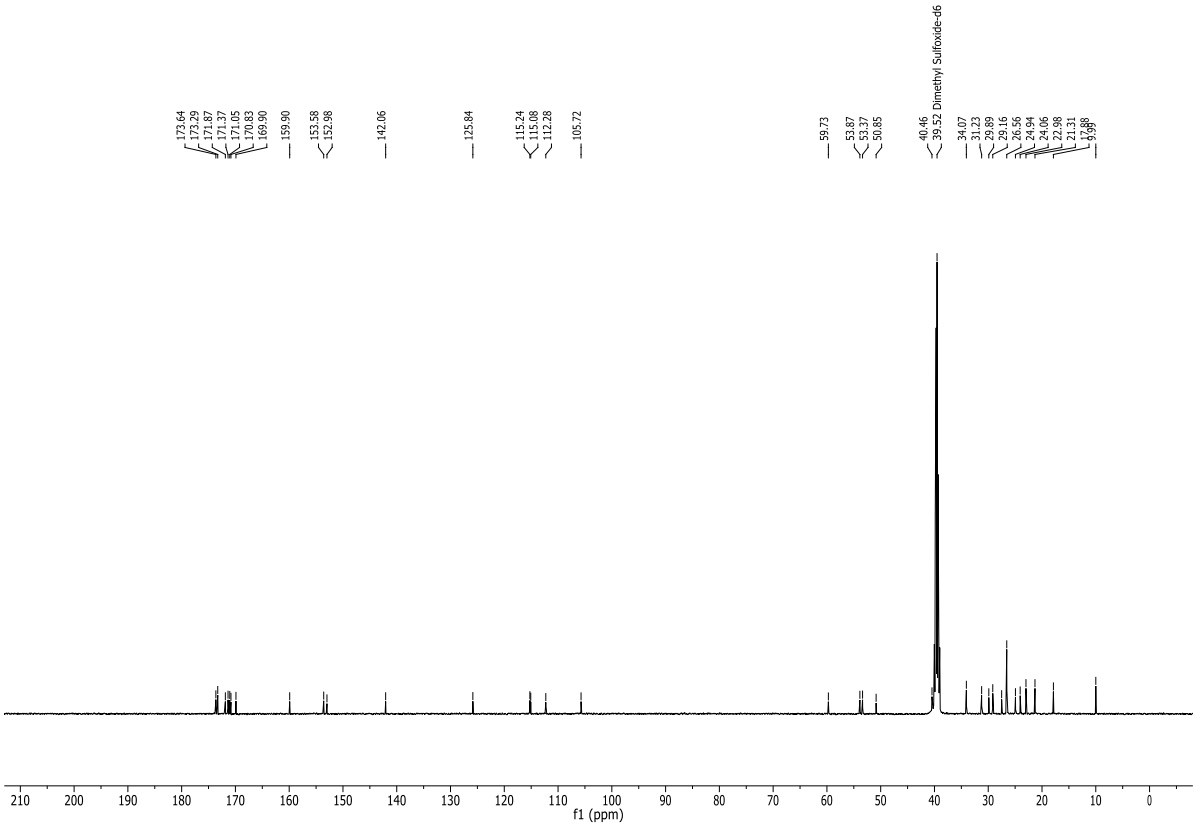
4.4.2. ^{1}H NMR and ^{13}C NMR Spectra of Compound 3





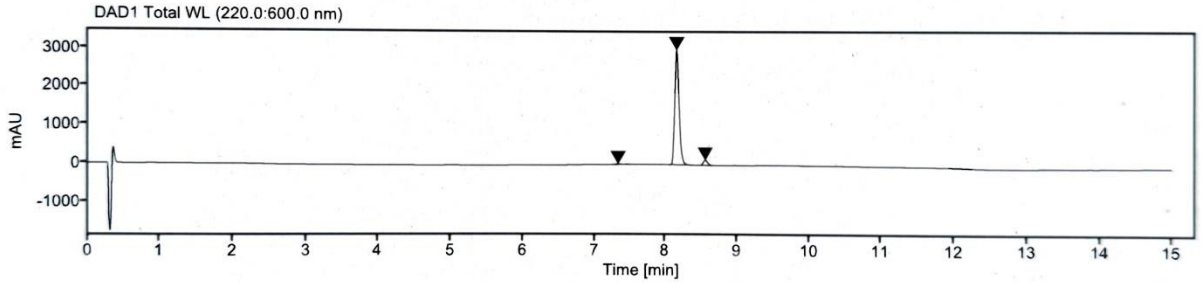
4.4.3. ^{1}H NMR and ^{13}C NMR Spectra of Compound 4





4.5. LC-MS Traces

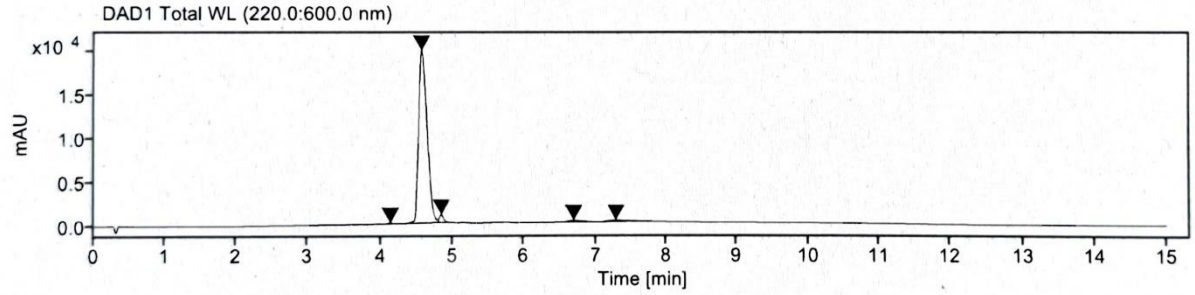
Compound **1**



Signal: DAD1 Total WL (220.0:600.0 nm)

RT [min]	Peak MS Base Peak m/z	Area	Area%	Max Peak%	Height
7.339		56.6582	0.4202	0.440	13.828
8.162		12872.0491	95.4596	100.000	3023.964
8.562		555.5801	4.1202	4.316	132.401
	Sum	13484.2875			

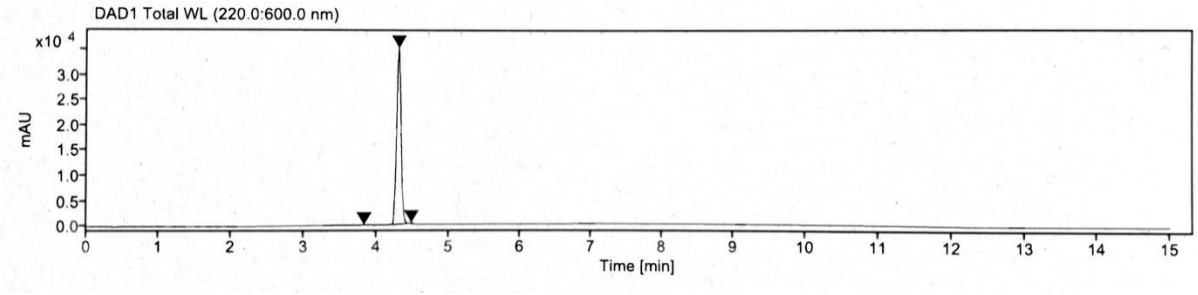
Compound **3**



Signal: DAD1 Total WL (220.0:600.0 nm)

RT [min]	Peak MS Base Peak m/z	Area	Area%	Max Peak%	Height
4.132		361.9427	0.2123	0.223	101.441
4.555		161976.3263	95.0294	100.000	19845.850
4.838		3629.8874	2.1296	2.241	885.081
6.688		2274.9988	1.3347	1.405	174.436
7.283		2205.5394	1.2940	1.362	140.254
	Sum	170448.6945			

Compound **4**



Signal: DAD1 Total WL (220.0:600.0 nm)

RT [min]	Peak MS Base Peak m/z	Area	Area%	Max Peak%	Height
3.831		132.6326	0.0861	0.086	40.467
4.303	713.500	153332.5439	99.5575	100.000	35063.426
4.484	715.600	548.8468	0.3564	0.358	185.964
	Sum	154014.0232			

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